

Mechanics of Solids (3130608)



Dedicated Faculty, Committed Education

Darshan
Institute of Engineering & Technology

Fundamentals of Statics



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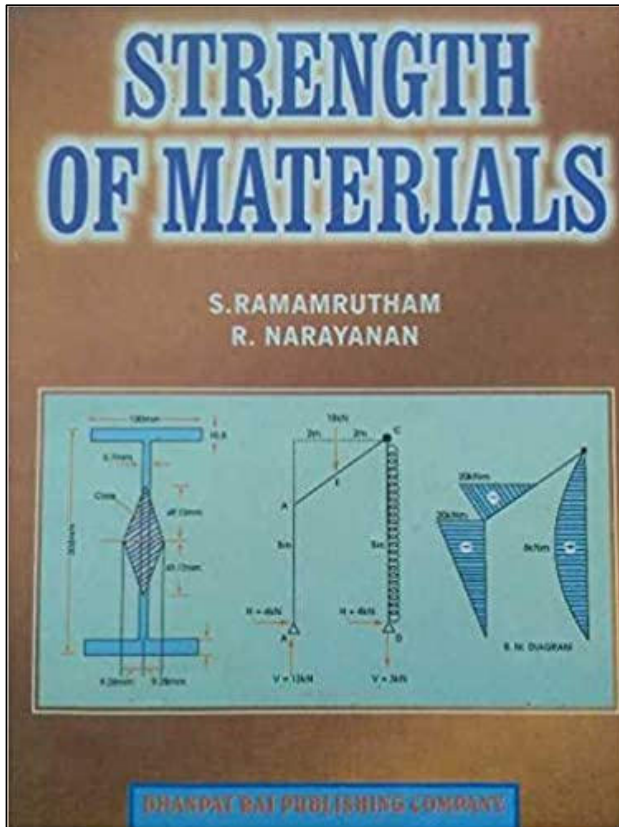
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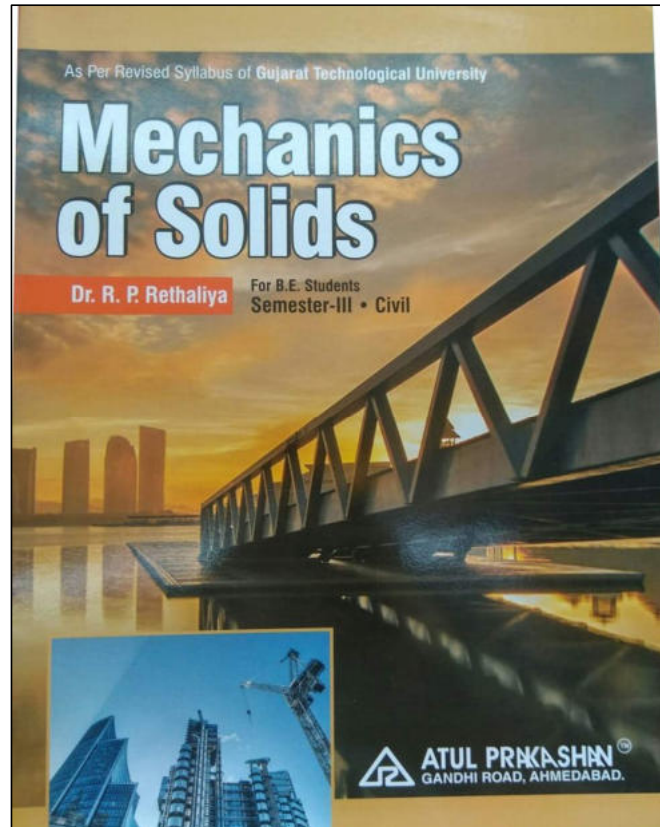
Contents of Presentation

- ▶ Reference Book and Text Book of Subject
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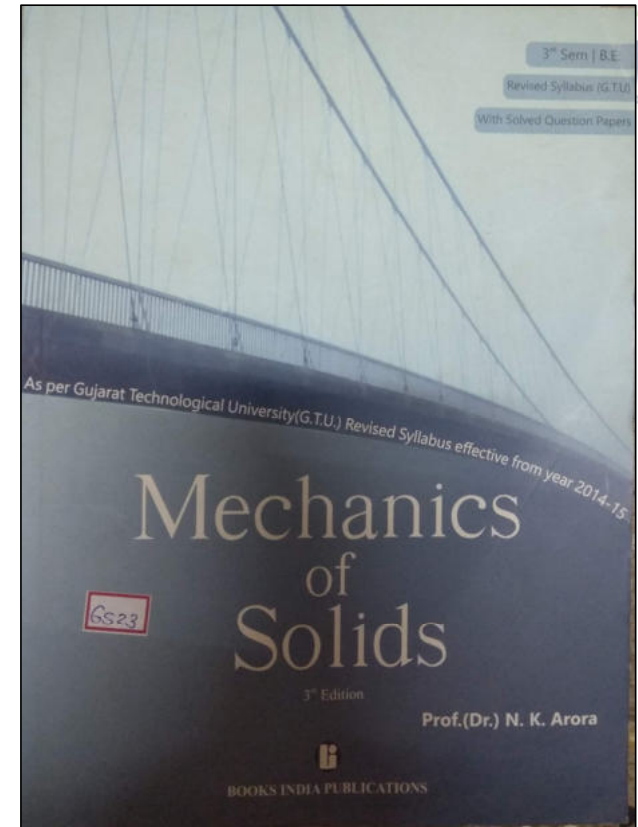
Reference Book and Text Book of Subject



Reference Book



Text Book



Text Book

Syllabus – Module 1 Description



GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering

Subject Code: 3130608

Semester III

Mechanics of Solids

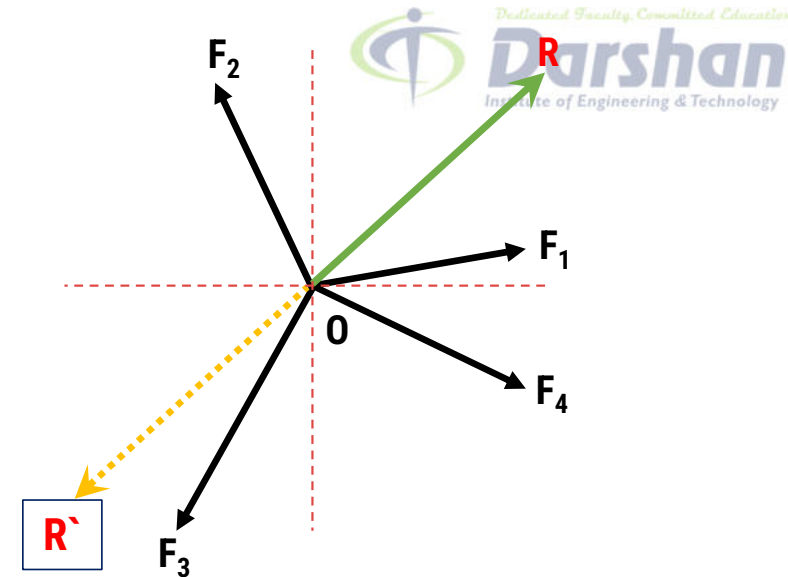
MODULE 1

Covered in Previous Lecture

1	Introduction Definition of space, time, particle, rigid body, deformable body. Force, types of forces, Characteristics of a force, System of forces, Composition and resolution of forces. Fundamental Principles of mechanics: Principle of transmissibility, Principle of superposition, Law of gravitation, Law of parallelogram of forces, Newton's Laws of Motion	02	25
2	Fundamentals of Statics Coplanar concurrent and non-concurrent force system: Resultant, Equilibrant, Free body diagrams. Coplanar concurrent forces: Resultant of coplanar concurrent force system by analytical and graphical method, Law of triangle of forces, Law of polygon of forces, Equilibrium conditions for coplanar concurrent forces, Lami's theorem. Application of these principles.	12	

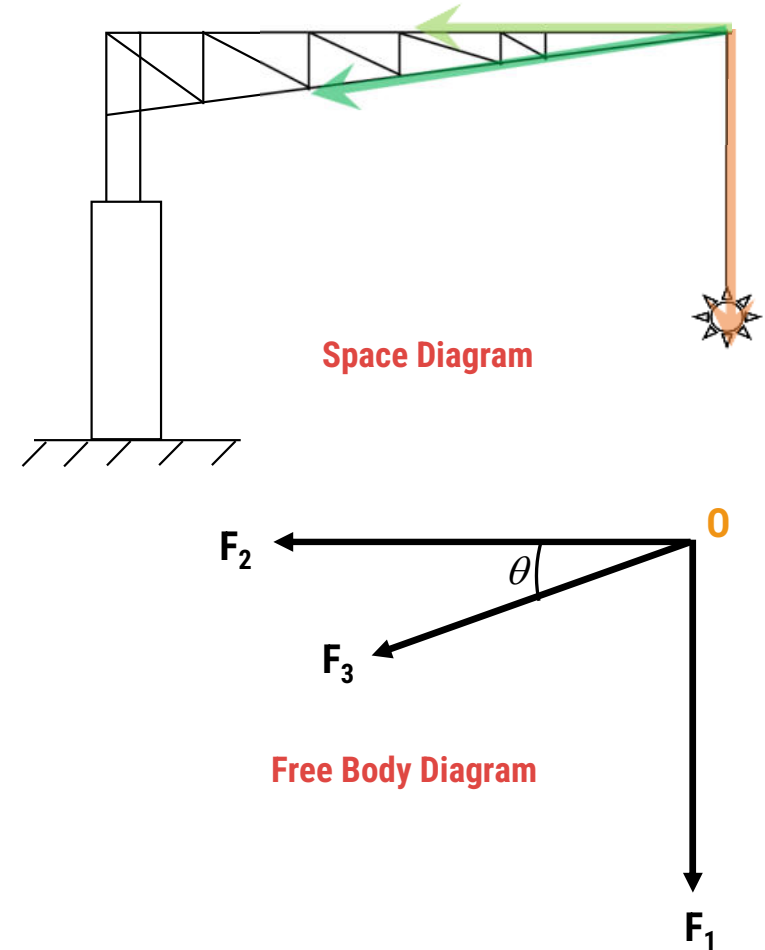
Definition : Resultant and Equilibrant

- ▶ **Resultant** : A force system acting on a body can be replaced by a single force, which produces same effect as that of the entire force system, is called the resultant force of the system
- ▶ **Equilibrant** : If a body is acted by a force system, body will move in the direction of resultant. If an equal & opposite force is applied in same line of action of resultant, the body will become stationary & system will be in equilibrium. This force is known as equilibrant
- ▶ As shown in figure $F_1 - F_4$ are acting at "O" point. "R" is resultant of all forces and "**R`**" is called as Equilibrant



Definition : Space Diagram & Free body diagram

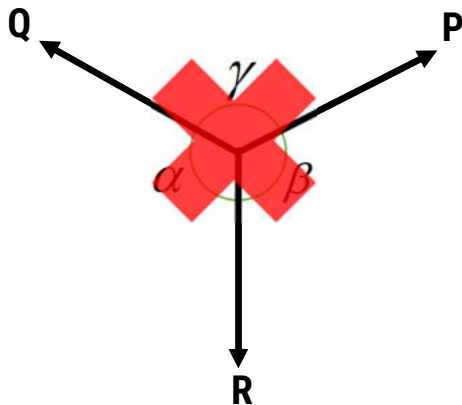
- ▶ **Free body diagram** : A significant point is selected from space diagram (actual problem) and drawing a separate diagram showing this significant point and all the forces acting on it, such diagram is called free body diagram.
- ▶ **Space diagram** : A sketch showing the physical conditions of the problem is known as a space diagram.



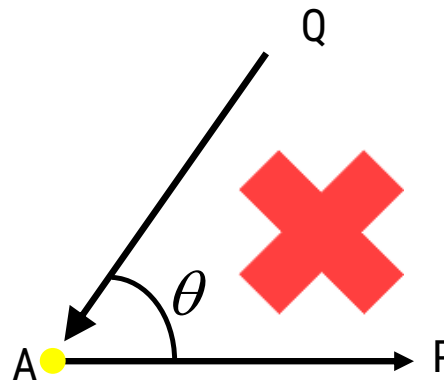
Law of Parallelogram of Forces

Condition for it's application

No. of forces are two only

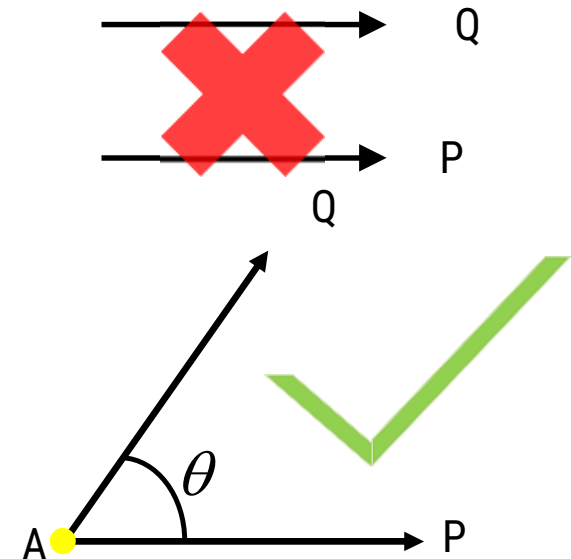


Nature of forces should be same



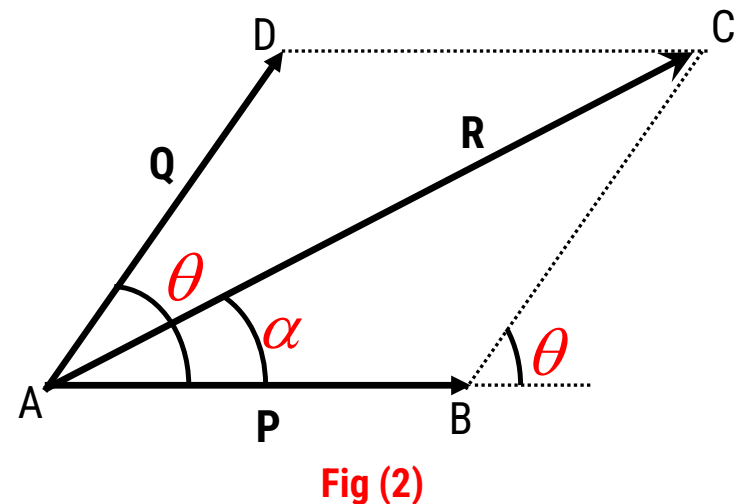
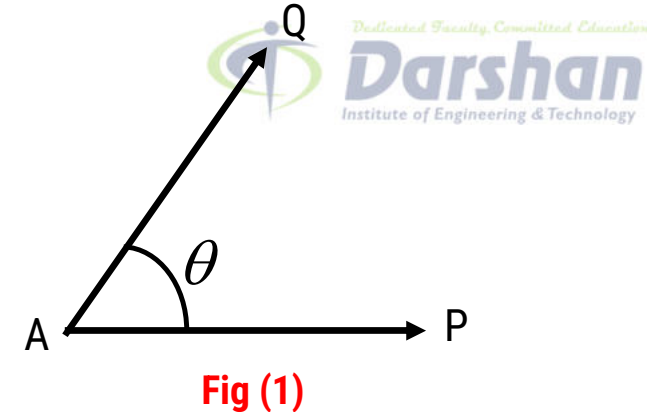
Concurrent coplanar forces

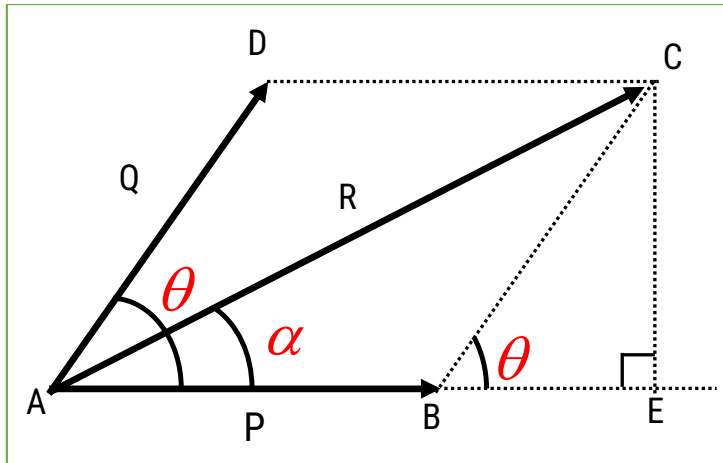
(Forces are passing through single point)



Law of Parallelogram of Forces

- ▶ If two concurrent forces are acting at a point on a rigid body & if these two forces are represented by two adjacent sides of a parallelogram then the diagonal passing through these two forces represents magnitude & direction of resultant force
- ▶ Let two forces P & Q acting on a particle as shown in fig.1 having angle between two forces is θ . The force P is represented as side AB with magnitude & direction & force Q is represented as an adjacent side AD of a parallelogram $ABCD$ as shown in fig. 2 the angle between AB & AD is θ , then the diagonal AC passing through point A represents the resultant R of the two forces P & Q & makes an angle α with force P .





▶ As shown in fig. drop a line from point C, perpendicular to line AB, let it intersect the line AB at E consider the Δ AEC,

▶ Here, AB = Force P
AD = Force Q
AC = Resultant R

Also, $AE = AB + BE$
 $BE = Q \cos \theta$
 $CE = Q \sin \theta$

▶ Applying rule of triangle for diagonal

$$AC^2 = AE^2 + CE^2$$

$$AC^2 = (AB + BE)^2 + CE^2$$

$$AC^2 = AB^2 + 2 \times AB \times BE + BE^2 + CE^2$$

$$R^2 = P^2 + 2PQ \cos \theta + Q^2 \cos^2 \theta + Q^2 \sin^2 \theta$$

$$R^2 = P^2 + 2PQ \cos \theta + Q^2 (\cos^2 \theta + \sin^2 \theta)$$

$$R = \sqrt{P^2 + 2PQ \cos \theta + Q^2}$$

.....magnitude of resultant

▶ Again consider the Δ AEC,

$$\tan \alpha = \frac{CE}{AE} = \frac{CE}{AB + BE}$$

$$= \frac{Q \sin \theta}{P + Q \cos \theta}$$

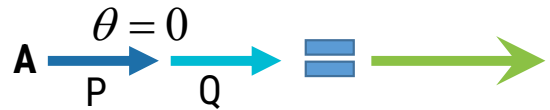
$$\alpha = \tan^{-1} \left[\frac{Q \sin \theta}{P + Q \cos \theta} \right]$$

.....direction of resultant

Law of Parallelogram of Forces for Special Cases

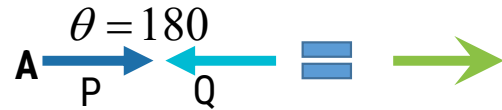
Case – 1

If two forces are acting parallel to each other in same direction


$$A \xrightarrow[\text{P}]{\theta = 0} \xrightarrow[\text{Q}]{} = \xrightarrow{\text{R}} \quad R = P + Q$$

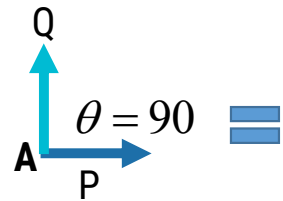
Case – 2

If two force are parallel to each other but in opposite direction, $\theta = 180$


$$A \xrightarrow[\text{P}]{\theta = 180} \xrightarrow[\text{Q}]{} = \xrightarrow{\text{R}} \quad R = P - Q$$

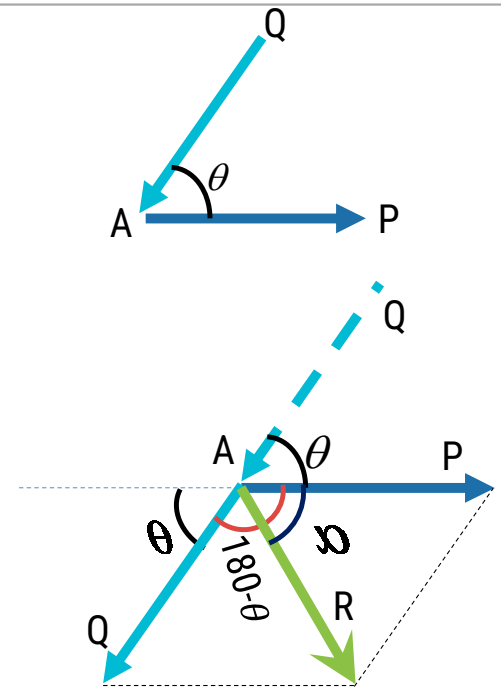
Case – 3

If two forces are perpendicular to each other, $\theta = 90$


$$A \xrightarrow[\text{P}]{\theta = 90} \xrightarrow[\text{Q}]{} = \xrightarrow{\text{R}} \quad R = \sqrt{P^2 + Q^2}$$

Case – 4

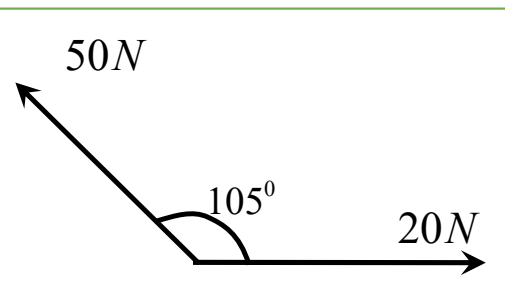
If nature of two forces different



Numerical on “Law of Parallelogram of Forces”

Ex - 1

Determine the magnitude and direction of the resultant of the two forces shown in figure



➤ **Answer**

From a given data

➤ By law of parallelogram

$$P = 20N \quad Q = 50N \quad \theta = 105^\circ$$

➤ Magnitude of resultant R is given by

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

➤ Where $P=20N$, $Q=50N$ and angle between both force is 105°

$$R = \sqrt{20^2 + 50^2 + 2 \times 20 \times 50 \cos 105}$$

$$R = 48.81N \quad \text{.....magnitude of resultant}$$

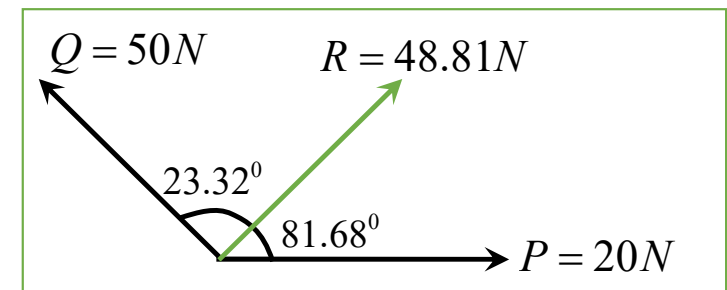
➤ Direction of resultant is given by

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$$

$$\alpha = \tan^{-1} \left(\frac{Q \sin \theta}{P + Q \cos \theta} \right)$$

$$\alpha = \tan^{-1} \left(\frac{50 \sin 105}{20 + 50 \cos 105} \right) = 81.68^\circ$$

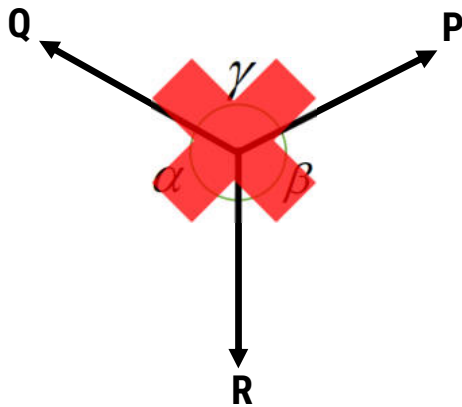
.....direction of resultant



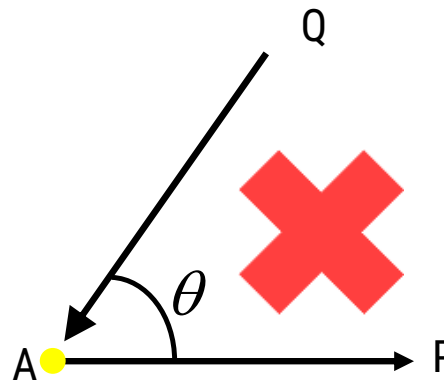
Law of “Triangle of Forces”

Condition for it's application

No. of forces are two only

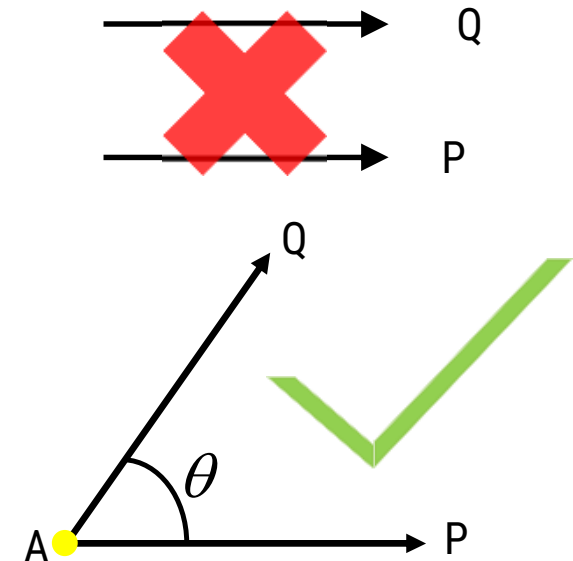


Nature of forces should be same



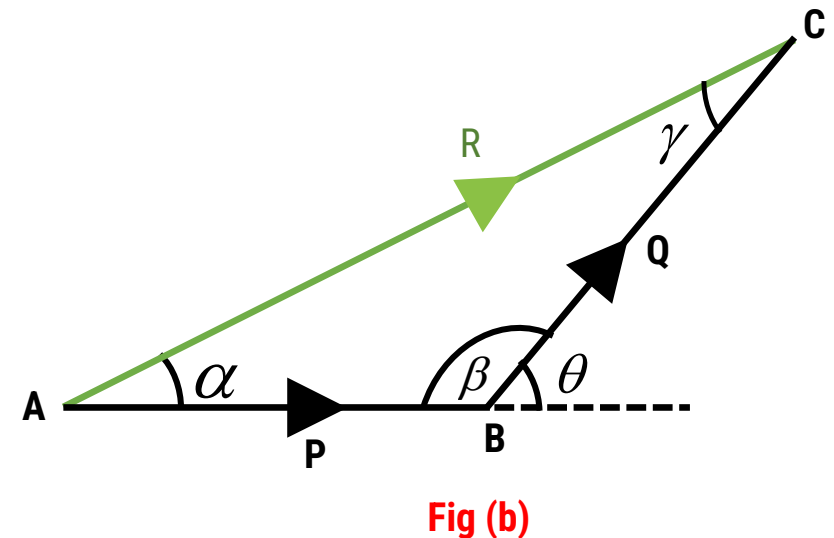
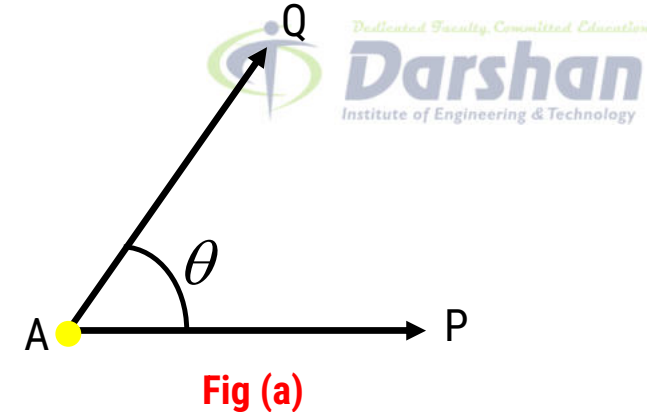
Concurrent Coplanar forces

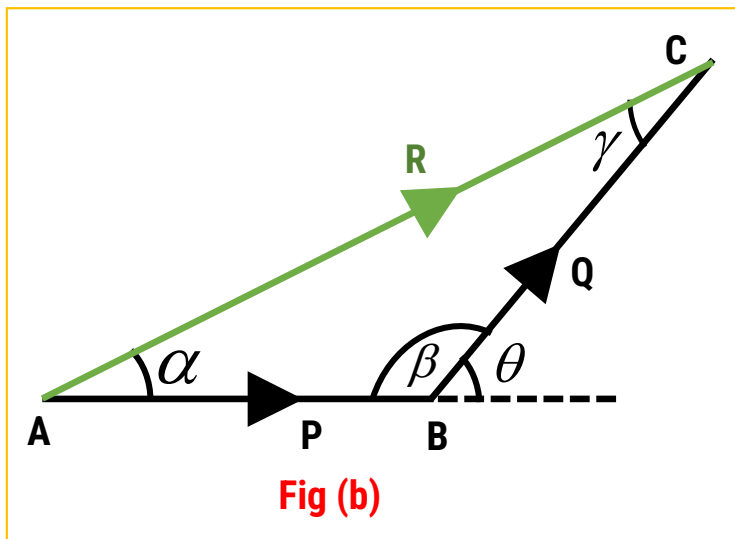
(Forces are passing through single point)



Law of “Triangle of Forces”

- ▶ Law of triangle state that, two concurrent forces are acting on a solid body, are represented as sides of a triangle with magnitude & direction in order than the third side of triangle gives the resultant in opposite order
- ▶ Let as shown in fig(a), P and Q are the two coplanar concurrent forces acting on a body at point A, at an angle θ .
- ▶ As shown in fig(b), according to the law of triangle, P and Q are taken as the sides of the triangle with, magnitude and direction in order, as sides AB and BC of a triangle, then the third side of triangle CA will represent the resultant R in opposite order as shown, α at a direction α with force P





- ▶ Here, AB = force P
BC = force Q

AC = resultant R
 $\beta = 180 - \theta$

- ▶ As shown in fig (B) applying cosine rule of triangle,

$$AC^2 = AB^2 + BC^2 - 2 \times AB \times BC \cos \beta$$

$$R^2 = P^2 + Q^2 - 2 \times P \times Q \cos \beta$$

$$R = \sqrt{P^2 + Q^2 - 2 \times P \times Q \cos \beta}$$

- ▶ As shown in fig (B) applying sine rule of triangle, we have

$$BC \propto \sin \angle CAB$$

$$BC = k \times \sin \angle CAB$$

$$\frac{BC}{\sin \angle CAB} = k$$

$$AC \propto \sin \angle CBA$$

$$AC = k \times \sin \angle CBA$$

$$\frac{AC}{\sin \angle CBA} = k$$

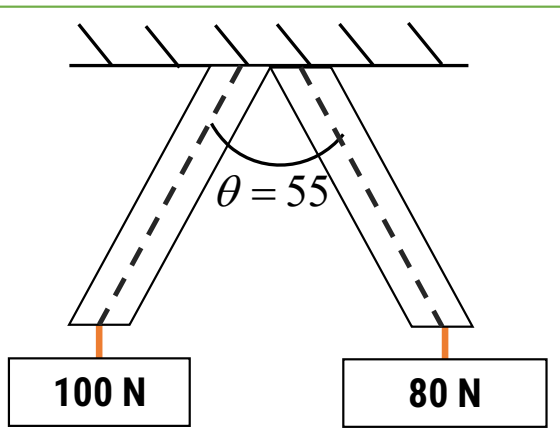
$$\frac{BC}{\sin \angle CAB} = \frac{AC}{\sin \angle CBA} \Rightarrow \frac{Q}{\sin \alpha} = \frac{R}{\sin \beta}$$

$$\sin \alpha = \left(\frac{Q}{R} \sin \beta \right) \Rightarrow \alpha = \sin^{-1} \left(\frac{Q}{R} \sin \beta \right)$$

Numerical on “Law of Triangle of Forces”

Ex - 2

Two buckets of weight 100 N and 80 N are suspended from a ceiling as shown in fig. find out reaction in supporting nail.



➤ **Answer**

From a given data

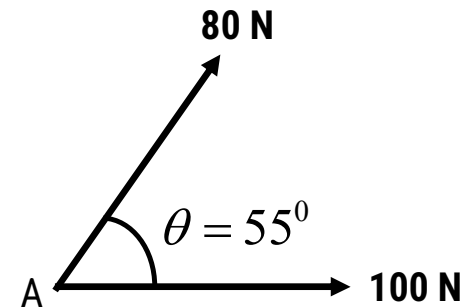
➤ Applying law of triangle as shown in fig

$$P = 100\text{ N}$$

$$Q = 80\text{ N}$$

$$\theta = 55^\circ$$

$$\beta = 180 - \theta = 125^\circ$$



➤ Magnitude of resultant R is given by

$$R = \sqrt{P^2 + Q^2 - 2 \times P \times Q \cos \beta}$$

$$R = \sqrt{100^2 + 80^2 - 2 \times 100 \times 80 \times \cos 125}$$

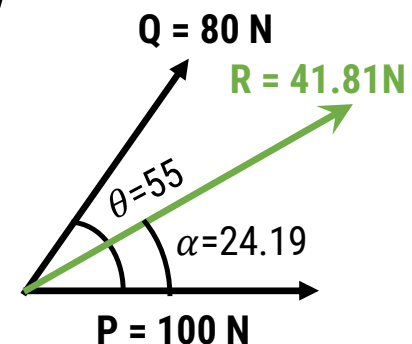
$$R = 159.93\text{ N} \text{magnitude of resultant}$$

➤ Direction of resultant is given by

$$\alpha = \sin^{-1} \left(\frac{Q}{R} \sin \beta \right)$$

$$\alpha = \sin^{-1} \left(\frac{80}{159.93} \sin 125 \right)$$

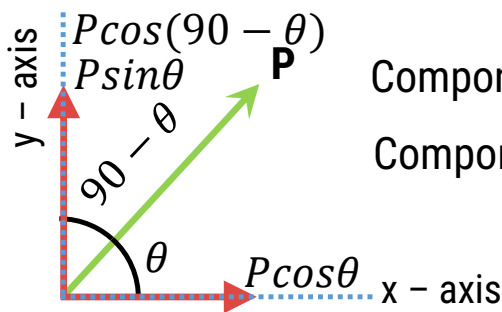
$$\alpha = 24.19^\circ \text{direction of resultant}$$



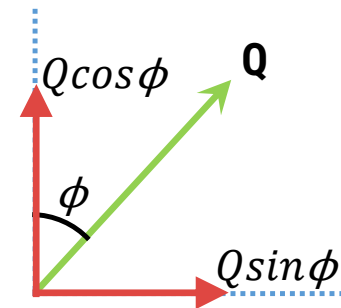
Resolution of a Force

- ▶ The process of splitting up the given force into a number of components, without changing its effect on the body is called resolution of a force. A force is, generally resolved along two mutually perpendicular directions.

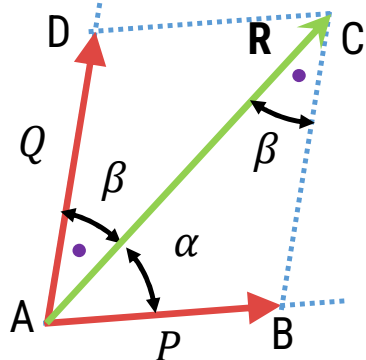
a) Orthogonal Resolution of forces or Perpendicular Components



Component of force along x - axis = $P \cos \theta$
 Component of force along y - axis = $P \cos (90 - \theta)$
 = $P \sin \theta$



b) Non-Orthogonal Resolution of forces or Non-Perpendicular Components



$AB = CD = P$
 $BC = AD = Q$
 $AC = R$

Applying sine rule to $\triangle ABC$

$$\frac{AB}{\sin \angle ACB} = \frac{BC}{\sin \angle CAB} = \frac{AC}{\sin \angle ABC}$$

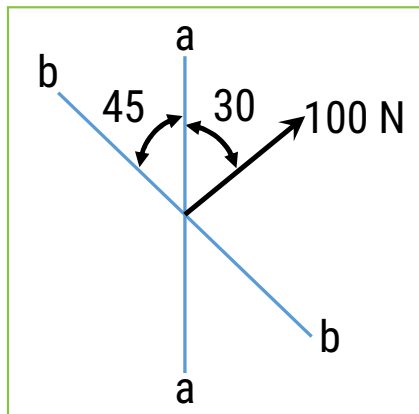
$$\frac{P}{\sin \beta} = \frac{Q}{\sin \alpha} = \frac{R}{\sin(\alpha + \beta)}$$

$$P = \frac{\sin \beta}{\sin(\alpha + \beta)} \cdot R$$

$$Q = \frac{\sin \alpha}{\sin(\alpha + \beta)} \cdot R$$

Numerical on Resolution of a Force

Ex - 3 Resolve 100 N force as shown in fig. along a – a and b – b



➤ **Answer** From a given data

Given Force = 100 N

➤ Split a force in to a given axis "a – a" & "b – b"

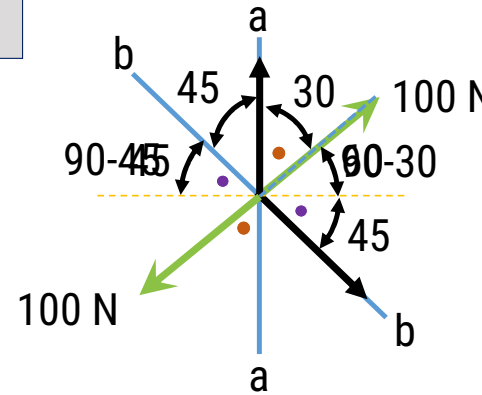


Fig (a)

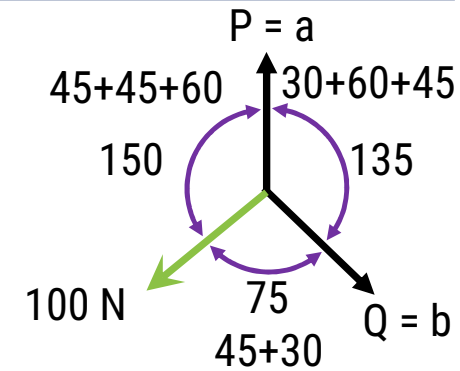


Fig (b)

➤ Applying sine rule in fig.(b)

$$\frac{P}{\sin 75} = \frac{Q}{\sin 150} = \frac{100}{\sin 135} \Rightarrow \frac{P}{0.966} = \frac{Q}{0.5} = \frac{100}{0.707}$$

$$P = 0.966 \times \frac{100}{0.707} = 136.63 \text{ N}$$

➤ Similarly, for Q force

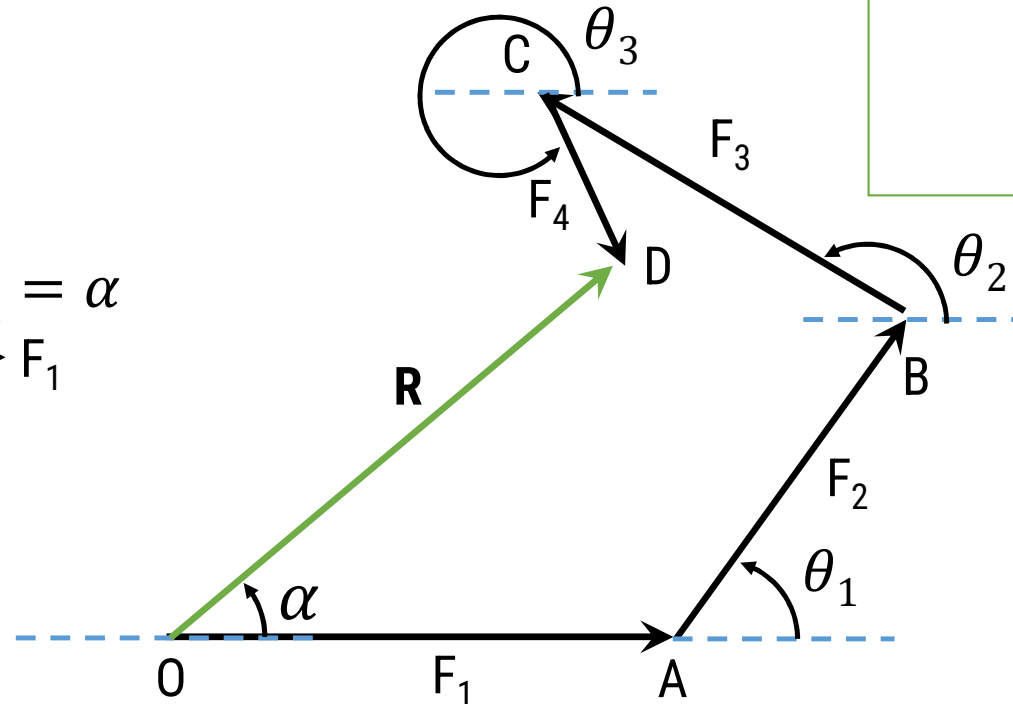
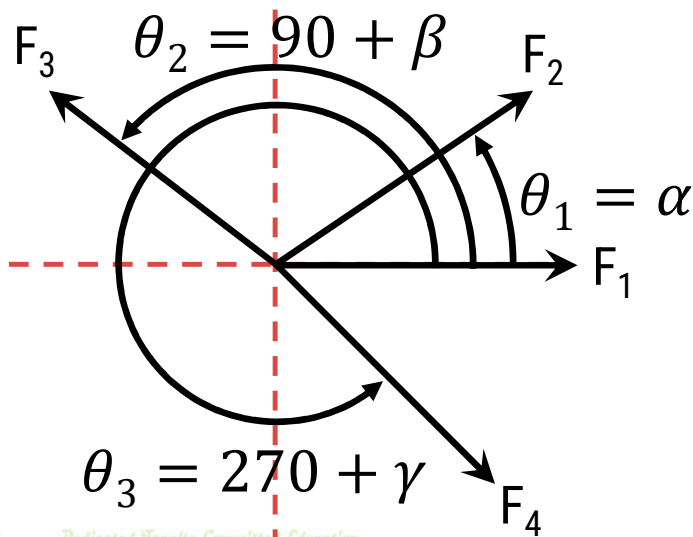
$$Q = 0.5 \times \frac{100}{0.707} = 70.72 \text{ N}$$

➤ Component along a-a = P = 136.63 N

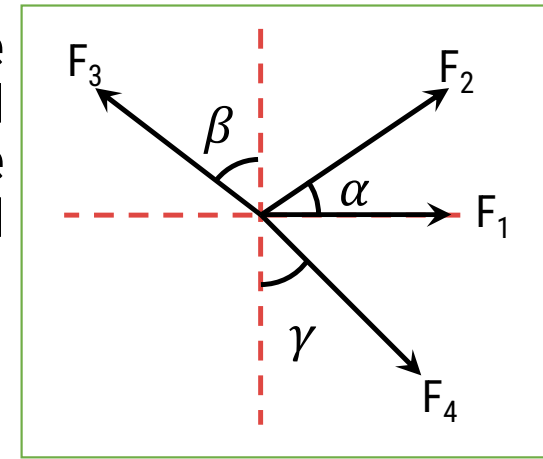
➤ Component along b-b = Q = 70.72 N

Law of Polygon of Forces

- ▶ When more than two coplanar forces acting at a point are represented by the sides of a polygon taken in order, in direction and magnitude, the closing line from first point to the last point of the polygon represents the resultant of the force system in direction and magnitude.

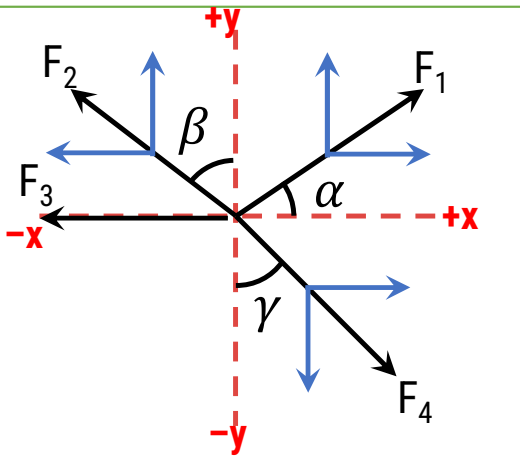


$$\begin{aligned} OA &= F_1 \\ AB &= F_2 \\ BC &= F_3 \\ CD &= F_4 \\ OD &= R \end{aligned}$$



Law of Polygon – Analytical Method No. 01

Determine the resultant of force system as shown in fig. by analytically

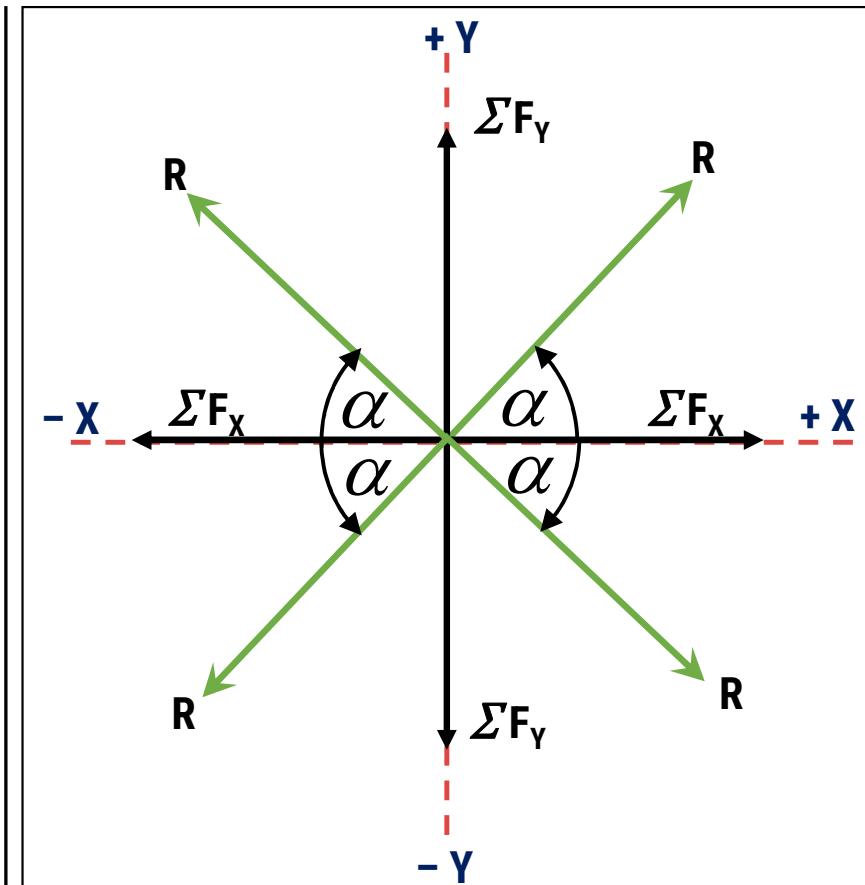


➤ Answer From a given data

Force	Angle
F_1	$\theta_1 = \alpha$
F_2	$\theta_2 = \beta$
F_3	$\theta_3 = 180$
F_4	$\theta_4 = \gamma$

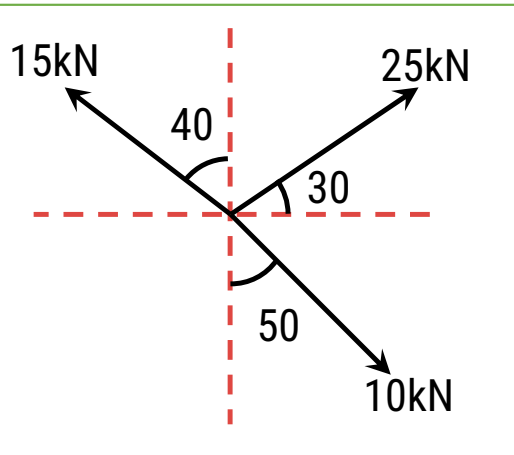
➤ Direction of Resultant with reference to positive x - axis

ΣF_x	+	—	—	+
ΣF_y	+	+	—	—
θ_R	α	$180 - \alpha$	$180 + \alpha$	$360 - \alpha$



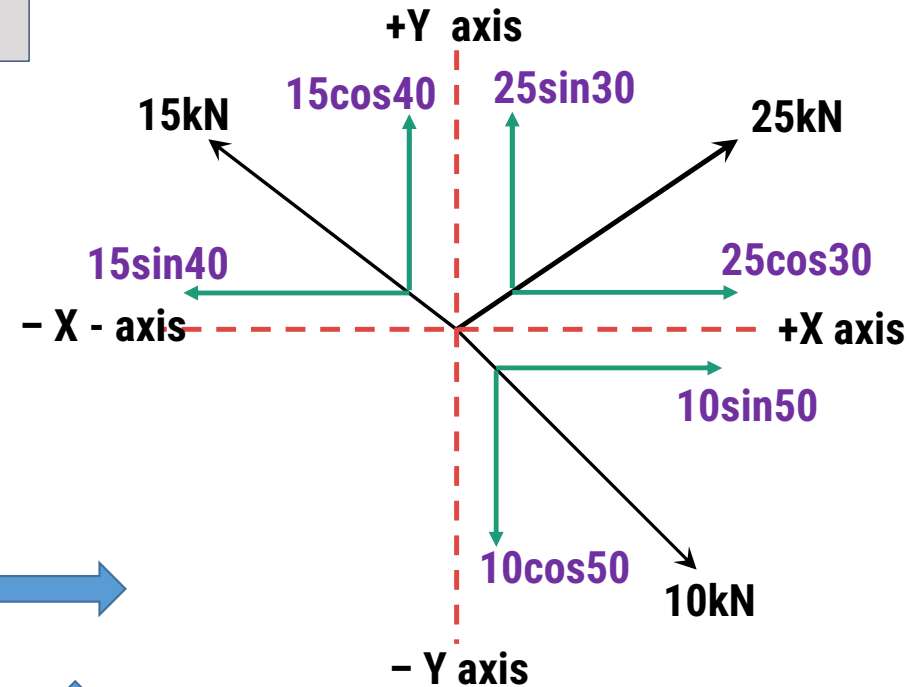
Numerical on “Law of Polygon of Forces”

Ex - 4 Determine the resultant of force system as shown in fig. by analytically



➤ **Answer** From a given data

Force	Angle
$F_1 = 25 \text{ kN}$	$\theta_1 = 30$
$F_2 = 15 \text{ kN}$	$\theta_2 = 40$
$F_3 = 10 \text{ kN}$	$\theta_3 = 50$



➤ Summation of horizontal force

$$\sum F_x = 25 \cos 30 - 15 \sin 40 + 10 \sin 50 = 19.67 \text{ kN} \rightarrow$$

➤ Summation of Vertical force

$$\sum F_y = 25 \sin 30 + 15 \cos 40 - 10 \cos 50 = 17.56 \text{ kN} \uparrow$$

$$\Sigma F_X = 19.67kN \rightarrow \quad \Sigma F_Y = 17.56kN \uparrow$$

► Magnitude of Resultant

$$R = \sqrt{(\Sigma F_X)^2 + (\Sigma F_Y)^2}$$

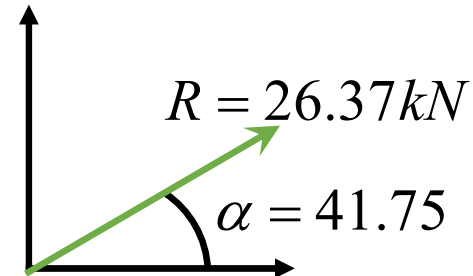
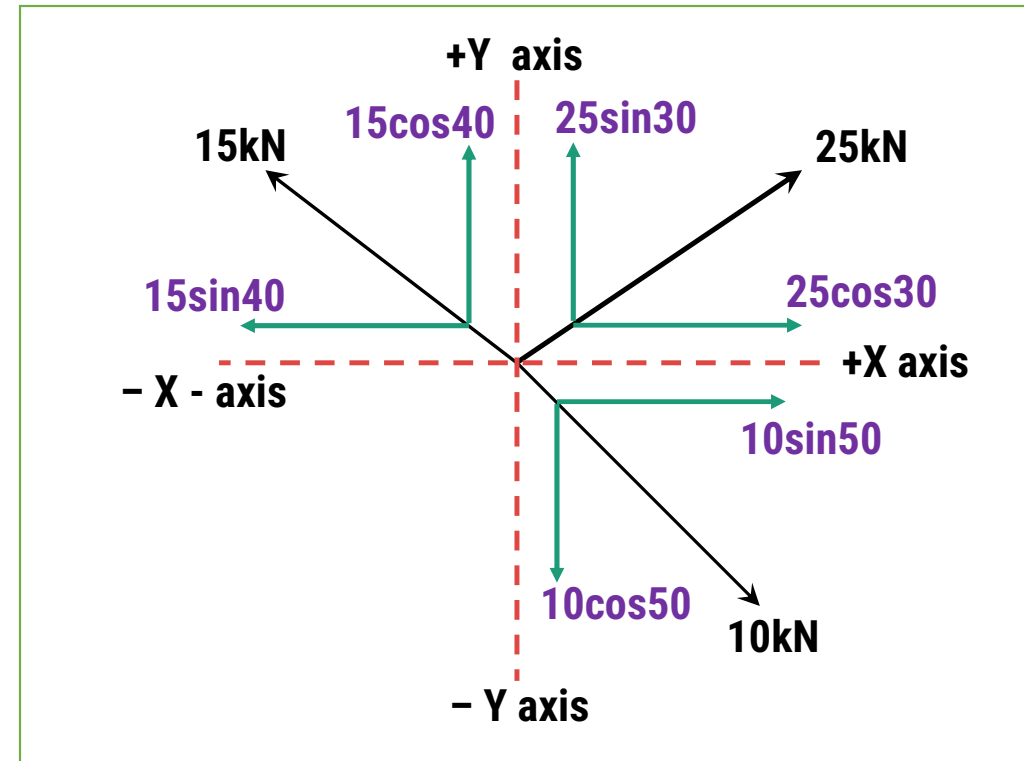
$$R = \sqrt{(19.67)^2 + (17.56)^2}$$

$$R = 26.37kN$$

► Direction of Resultant with reference to positive x - axis

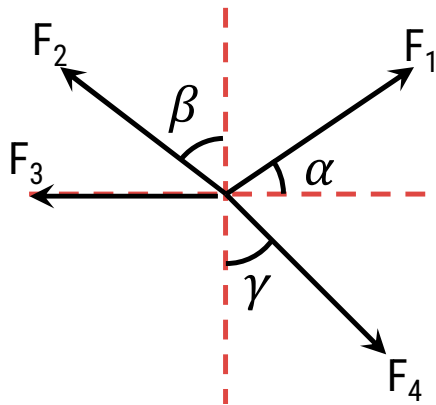
$$\tan \alpha = \frac{\Sigma F_Y}{\Sigma F_X} = \frac{17.56}{19.67} = 0.89$$

$$\alpha = \tan^{-1}(0.89) = 41.75^\circ$$



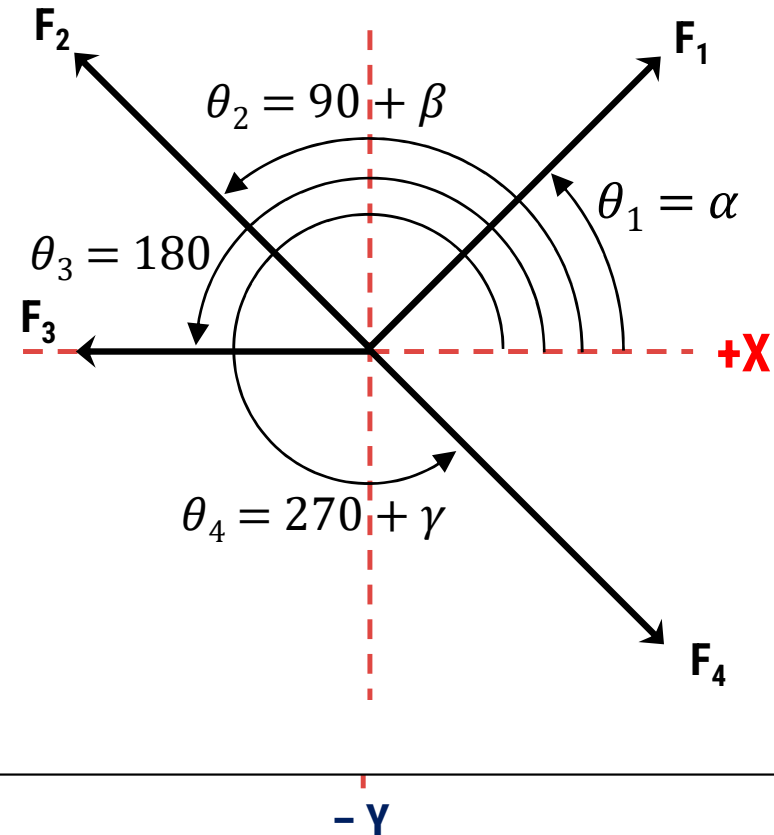
Law of Polygon – Analytical Method No. 02

Determine the resultant of force system as shown in fig. by analytically



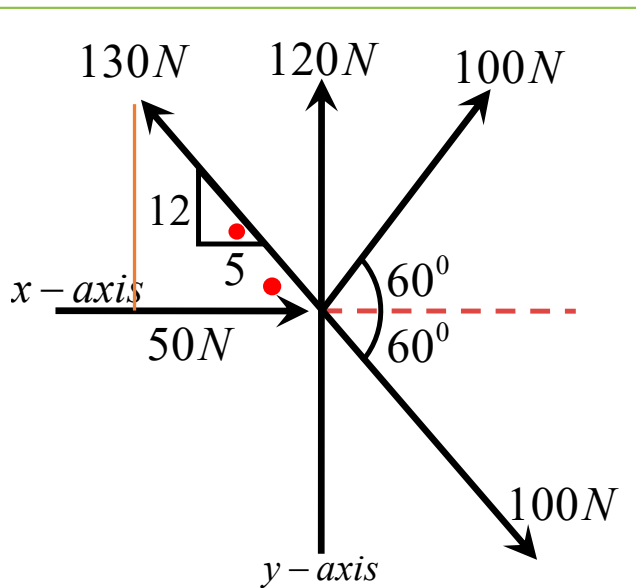
➤ Answer From a given data

Force	Angle
F_1	$\theta_1 = \alpha$
F_2	$\theta_2 = 90 + \beta$
F_3	$\theta_3 = 180$
F_4	$\theta_4 = 270 + \gamma$

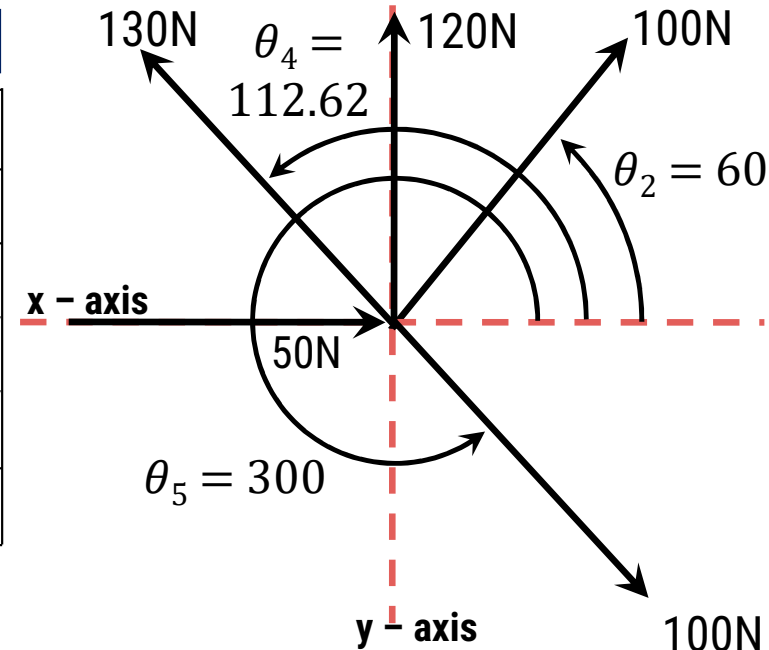


➤ Direction of Resultant with reference to positive x - axis

ΣF_x	+	—	—	+
ΣF_y	+	+	—	—
θ_R	α	$180 - \alpha$	$180 + \alpha$	$360 - \alpha$

Ex - 5**Determine the resultant of force system as shown in fig. by analytically****➤ Answer****From a given data**

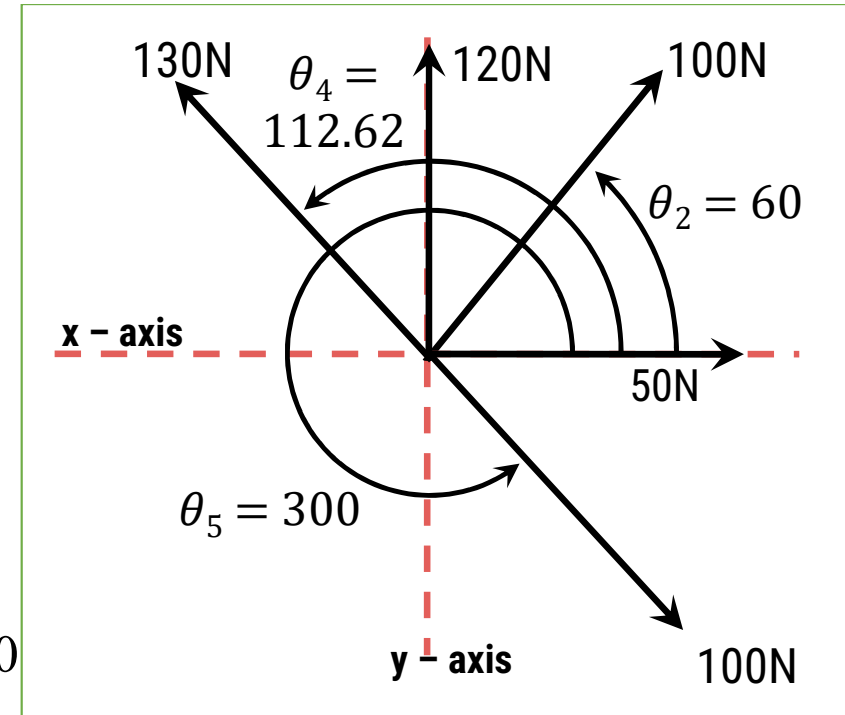
Force	Angle
$F_1 = 50\text{N}$	$\theta_1 = 0$
$F_2 = 100\text{N}$	$\theta_2 = 60$
$F_3 = 120\text{N}$	$\theta_3 = 90$
$F_4 = 130\text{N}$	$\theta_4 = 112.62$
$F_5 = 100\text{N}$	$\theta_5 = 300$

**► Summation of horizontal force**

$$\begin{aligned}\sum F_x &= F_1 \cos \theta_1 + F_2 \cos \theta_2 + \dots + F_n \cos \theta_n \\ &= 50 \cos 0 + 100 \cos 60 + 120 \cos 90 + 130 \cos 112.62 + 100 \cos 300 \\ \sum F_x &= 100\text{kN} \rightarrow\end{aligned}$$

$$\begin{aligned}\alpha &= \tan^{-1} \left(\frac{12}{5} \right) = 67.38 \\ \theta_4 &= 180 - 67.38 = 112.62^\circ\end{aligned}$$

Force	Angle
$F_1 = 50\text{N}$	$\theta_1 = 0$
$F_2 = 100\text{N}$	$\theta_2 = 60$
$F_3 = 120\text{N}$	$\theta_3 = 90$
$F_4 = 130\text{N}$	$\theta_4 = 112.62$
$F_5 = 100\text{N}$	$\theta_5 = 300$



► Summation of Vertical force

$$\sum F_Y = F_1 \sin \theta_1 + F_2 \sin \theta_2 + \dots + F_n \sin \theta_n$$

$$50 \sin 0 + 100 \sin 60 + 120 \sin 90 + 130 \sin 112.62 + 100 \sin 300$$

$$\sum F_Y = 240\text{kN} \uparrow$$

► Magnitude of Resultant

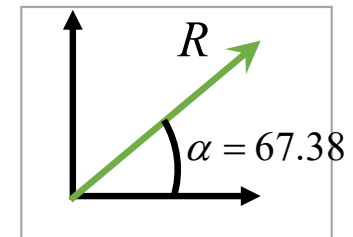
$$R = \sqrt{(\sum F_X)^2 + (\sum F_Y)^2} = 260\text{N}$$

► Direction of Resultant

$$\tan \alpha = \frac{\sum F_Y}{\sum F_X} = \frac{240}{100} = 67.38$$

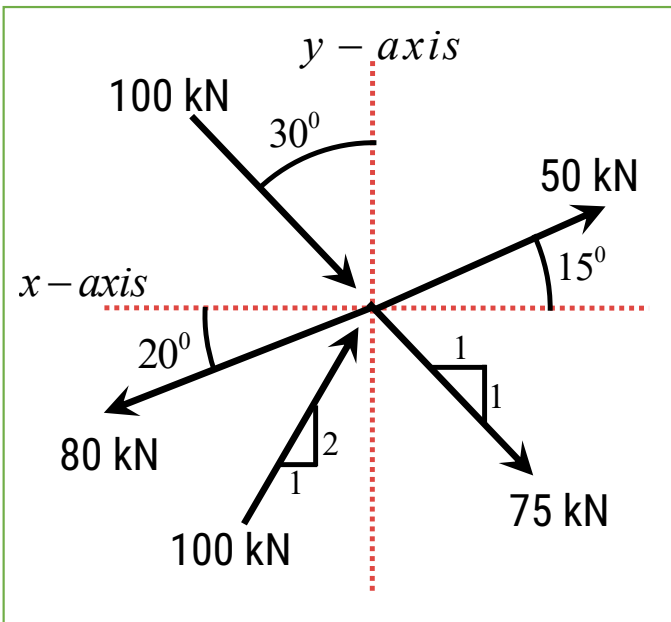
► Direction of Resultant with reference to positive x - axis

$$\theta_R = \alpha = 67.38$$



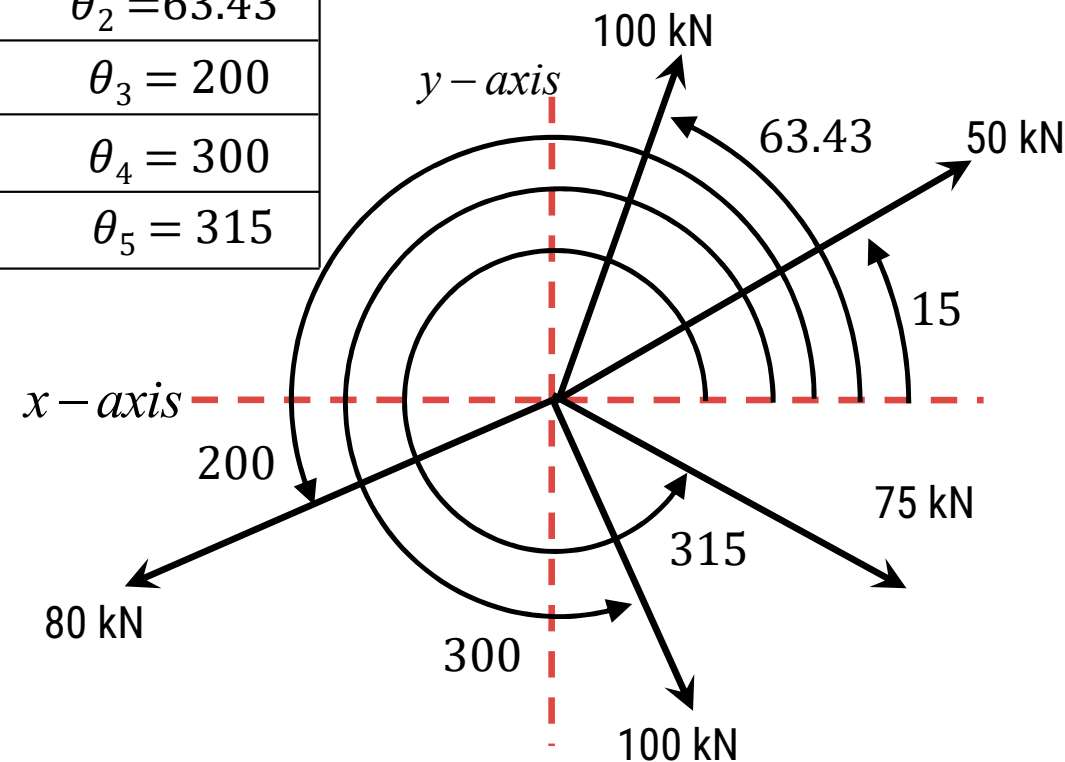
Ex - 6

Determine the resultant of force system as shown in fig. by analytically



➤ **Answer** **From a given data**

Force	Angle
$F_1 = 50 \text{ kN}$	$\theta_1 = 15$
$F_2 = 100 \text{ kN}$	$\theta_2 = 63.43$
$F_3 = 80 \text{ kN}$	$\theta_3 = 200$
$F_4 = 100 \text{ kN}$	$\theta_4 = 300$
$F_5 = 70 \text{ kN}$	$\theta_5 = 315$



► Angle between +x - axis and F_2 force

$$\alpha = \tan^{-1}\left(\frac{2}{1}\right) = 63.43 = \theta_2 = 63.43$$

Force	Angle
$F_1 = 50 \text{ kN}$	$\theta_1 = 15$
$F_2 = 100 \text{ kN}$	$\theta_2 = 63.43$
$F_3 = 80 \text{ kN}$	$\theta_3 = 200$
$F_4 = 100 \text{ kN}$	$\theta_4 = 300$
$F_5 = 70 \text{ kN}$	$\theta_5 = 315$

► Summation of horizontal force

$$\sum F_x = F_1 \cos \theta_1 + F_2 \cos \theta_2 + \dots + F_n \cos \theta_n$$

$$\sum F_x = 50 \cos 15 + 100 \cos 63.43 + 80 \cos 200 + 100 \cos 300 + 75 \cos 315$$

$$\sum F_x = 120.88 \text{ kN} \rightarrow$$

► Summation of vertical force

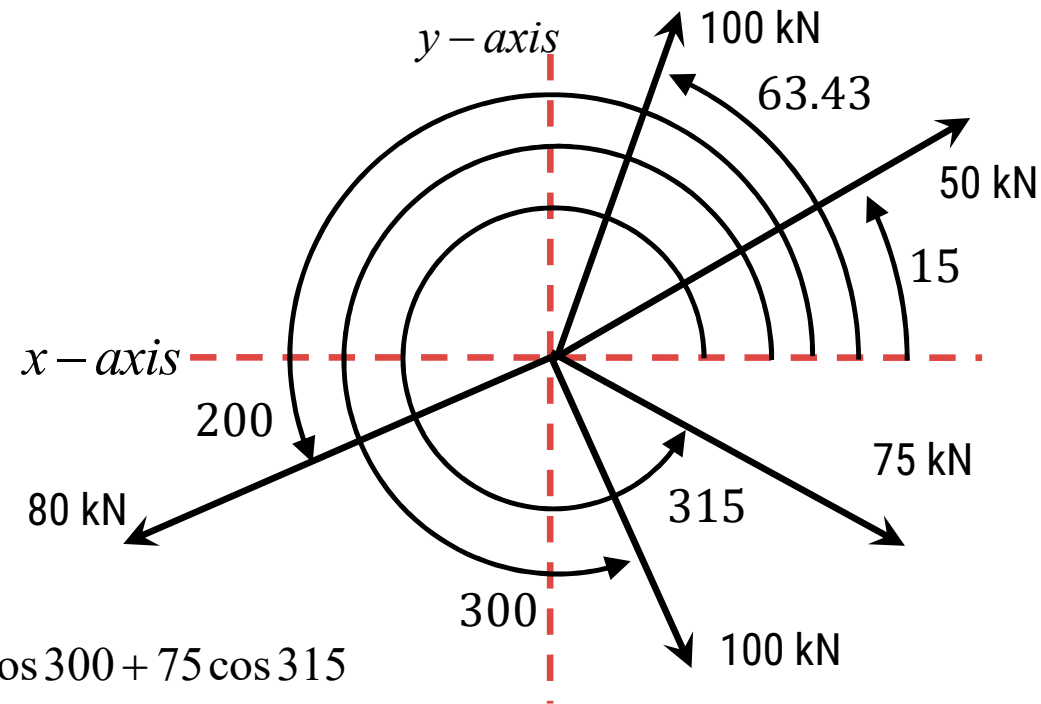
$$\sum F_y = F_1 \sin \theta_1 + F_2 \sin \theta_2 + \dots + F_n \sin \theta_n$$

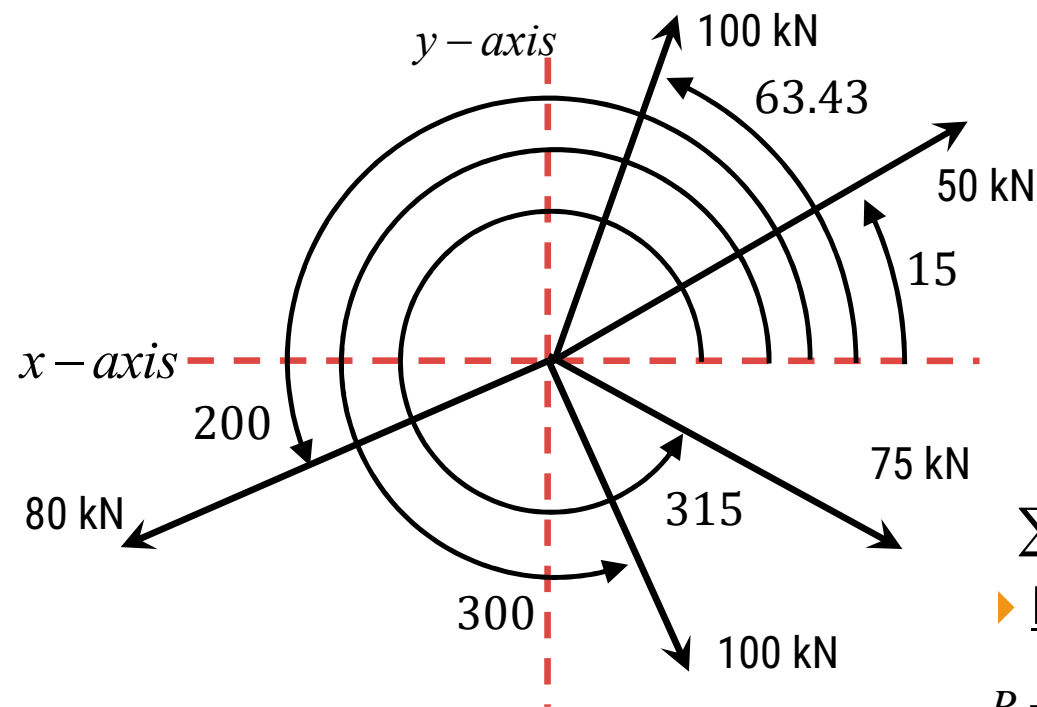
$$\sum F_y = 50 \sin 15 + 100 \sin 63.43 + 80 \sin 200 + 100 \sin 300 + 75 \sin 315$$

$$\sum F_y = -64.61 \text{ kN} \downarrow$$

$R = ?$

$\theta_R = ?$





Force	Angle
$F_1 = 50 \text{ kN}$	$\theta_1 = 15$
$F_2 = 100 \text{ kN}$	$\theta_2 = 63.43$
$F_3 = 80 \text{ kN}$	$\theta_3 = 200$
$F_4 = 100 \text{ kN}$	$\theta_4 = 300$
$F_5 = 70 \text{ kN}$	$\theta_5 = 315$

$R = ?$

$\theta_R = ?$

$$\sum F_x = 120.88 \text{ kN} \rightarrow \quad \sum F_y = -109.34 \text{ kN} \downarrow$$

► Magnitude of Resultant

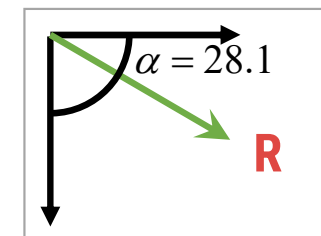
$$R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$$

$$R = \sqrt{(120.88)^2 + (-109.34)^2} = 137.06 \text{ kN}$$

► Direction of Resultant with reference to positive x - axis

$$\tan \alpha = \frac{\sum F_x}{\sum F_y} = \frac{64.61}{120.88} = 28.1$$

$$\theta_R = 360 - \alpha = 331.9$$



Lami's theorem

- ▶ When three concurrent coplanar forces are acting on a body and body is in equilibrium, then each force is directly proportion of the sine of the angle between the other two forces
- ▶ Let as shown in fig(a), three concurrent coplanar forces P, Q and R are acting on a body at point 'O' and body is in equilibrium.
- ▶ Applying Polygon's rules, taking forces P, Q and R as sides AB, BC and CA of the polygon, the system is in equilibrium, hence we have close polygon as shown in fig(b), a triangle will be obtained and applying sine rule

$$\frac{AB}{\sin \alpha} = \frac{BC}{\sin \beta} = \frac{CA}{\sin \gamma} \quad \Rightarrow \quad \frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$$

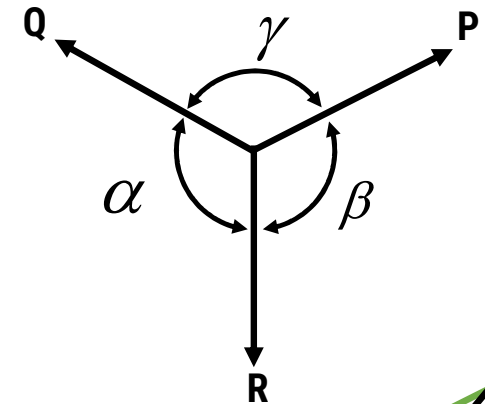


Fig (a)

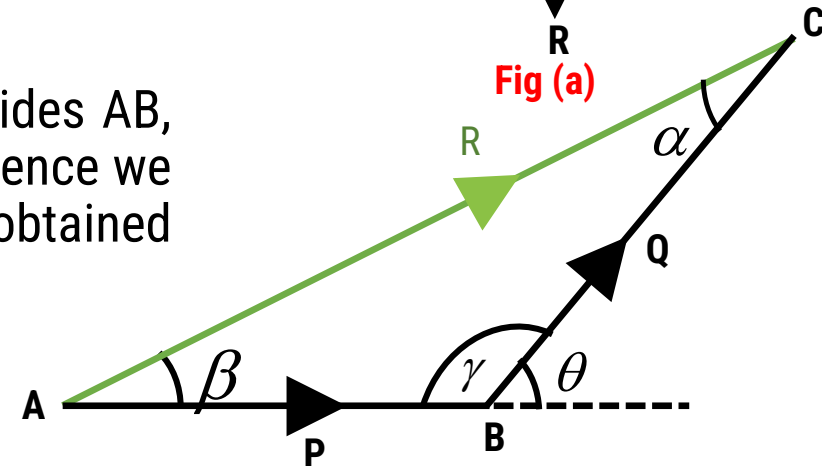


Fig (b)

Ex - 7

A cylindrical roller 600mm diameter and weighting 1000N is resting on a smooth inclined surface, tied firmly by a rope AC of length 600mm as shown in Fig. Find tension in rope and reaction at B.

➤ **Answer** **From a given data**

AB=600mm

BC=300mm...dia. of cylinder

Weight of cylinder is 1000N

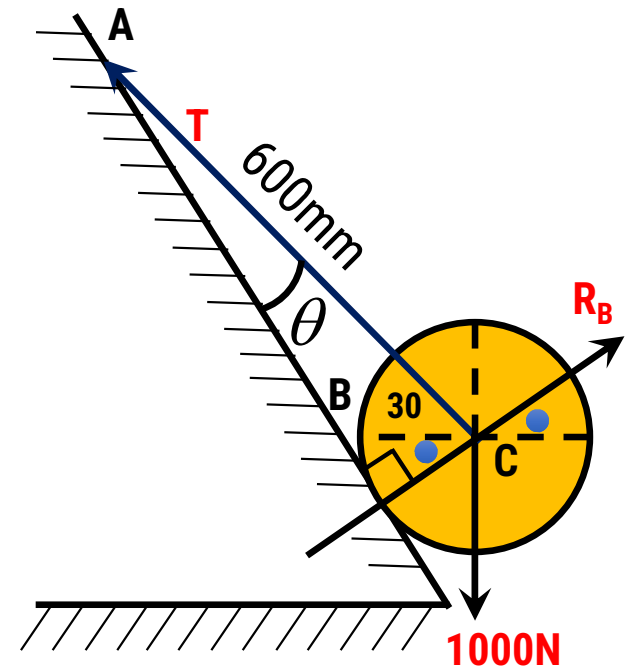
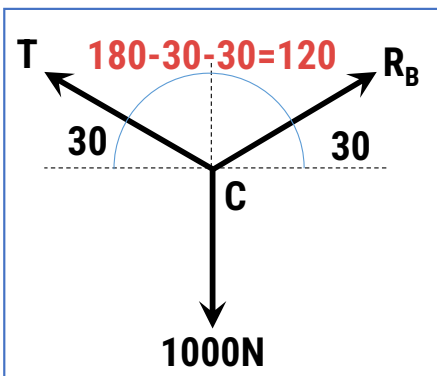
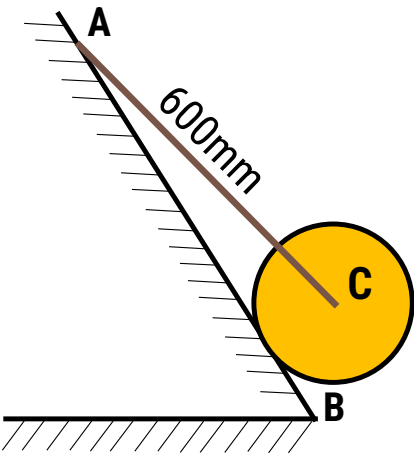
From geometry $\theta = \sin^{-1} \frac{300}{600} = 30$

Hence $\angle ABC = 90$ and $\angle ACB = 60$

▶ Applying Lami's theorem at Joint – C

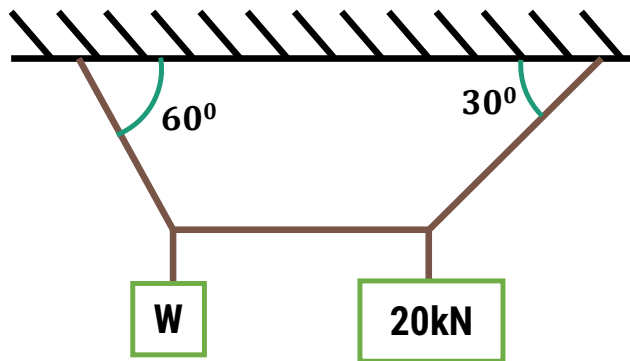
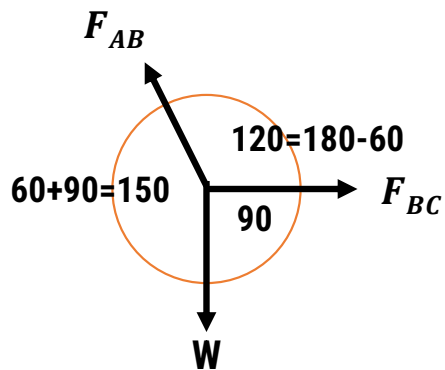
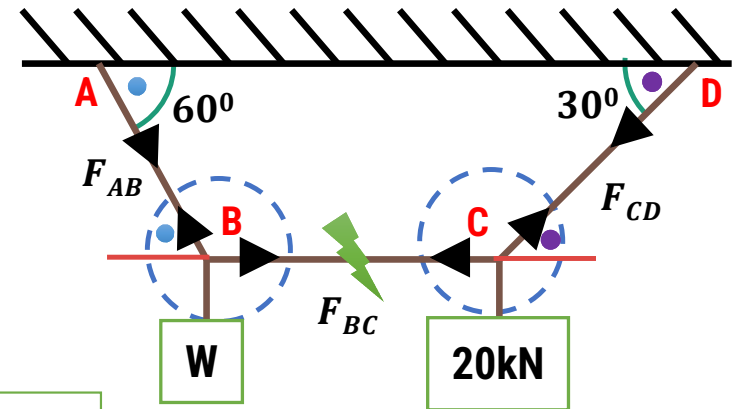
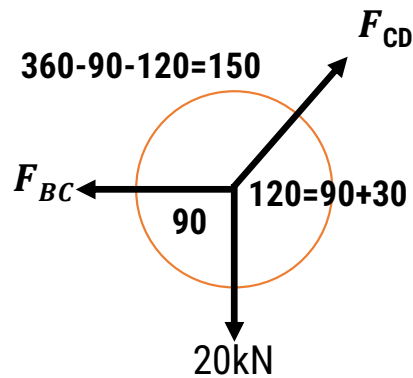
$$\frac{1000}{\sin 120} = \frac{R_B}{\sin 120} = \frac{T}{\sin 120}$$

▶ Hence, $R_B = T = 1000\text{N}$

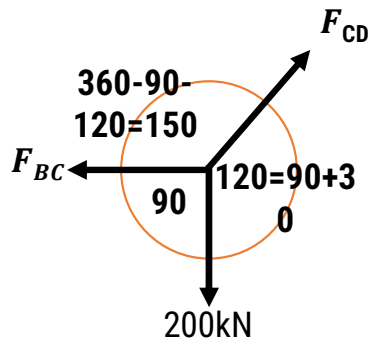


Ex - 8

A cord supported at B and C carries a load of 20 kN at C and a load of W at B as shown in figure. Find the value of W so that BC remains horizontal.

**Joint-B****Joint-C**

Joint-C

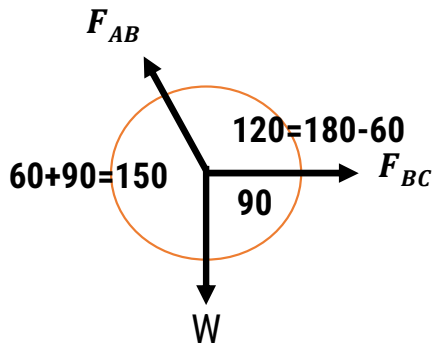


► Applying Lami's theorem at Joint – C

$$\frac{20}{\sin 150} = \frac{F_{BC}}{\sin 120} = \frac{F_{CD}}{\sin 90}$$

$$40 = \frac{F_{BC}}{\sin 120} = \frac{F_{CD}}{\sin 90} \quad \begin{cases} 40 \times \sin 120 = F_{BC} = 34.64kN \\ 40 \times \sin 90 = F_{CD} = 40kN \end{cases}$$

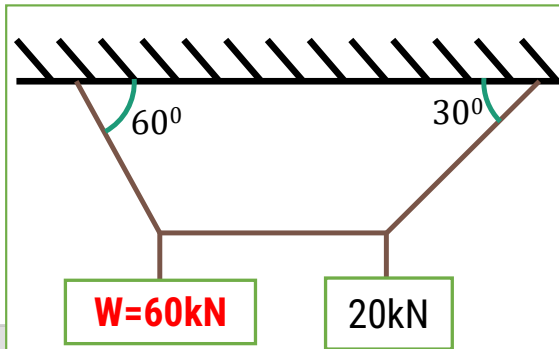
Joint-B



► Applying Lami's theorem at Joint – B

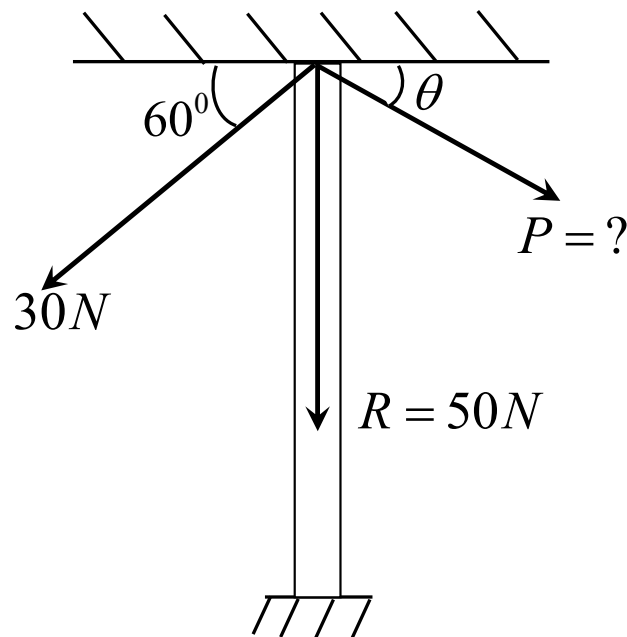
$$\frac{W}{\sin 120} = \frac{F_{BC} = 34.64}{\sin 150} = \frac{F_{AB}}{\sin 90}$$

$$\frac{W}{\sin 60} = 69.28 = \frac{F_{AB}}{\sin 90} \quad \begin{cases} 69.28 \times \sin 60 = W = 60kN \\ 69.28 \times \sin 90 = F_{AB} = 69.28kN \end{cases}$$



Ex - 9

The resultant force of a system of two forces is directed vertically downwards. Magnitude of force R is 50 N, one of the force of the system has magnitude of 30 N and is inclined at an angle of 60° with horizontal as shown in figure. Determine the magnitude of P and direction θ of the second force.

**➤ Answer****From a given data**

➤ Resultant is vertically downward, Hence

$$\sum F_x = 0$$

$$-30 \cos 60^\circ + P \cos \theta = 0$$

$$P \cos \theta = 15 \dots \text{eqn(1)}$$

$$R = \sum F_y$$

$$50 = P \sin \theta + 30 \sin 60$$

$$50 = P \sin \theta + 25.98$$

$$P \sin \theta = 24.02 \dots \text{eqn(2)}$$

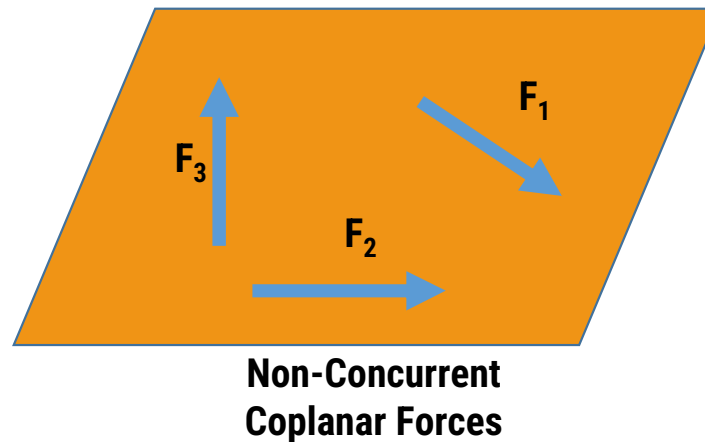
➤ From eqn (1) & (2)

$$P^2 \sin^2 \theta + P^2 \cos^2 \theta = P^2 = (15)^2 + (24.02)^2$$

$$P^2 = 801.9604 \quad \Rightarrow \quad P = 28.32 \text{ N}$$

Coplanar Non-Concurrent forces

- ▶ The forces, which do not meet at one point, but their lines of action lie on the same plane, are known as coplanar non-concurrent forces.



Definition : Moment

- ▶ **Moment of a force** : when a force acts on a particle, the particle moves or tends to move in the direction of the force, the motion being called as the motion of translation. However, if the force acts on a particle at some distance through an arm, it produces motion of rotation. That rotation is called as moment of a force.



Moment (M) =

Force(F) x $\perp$ distance(l)

$$M = F \times l$$

Type of moment

+ Ve (Positive)

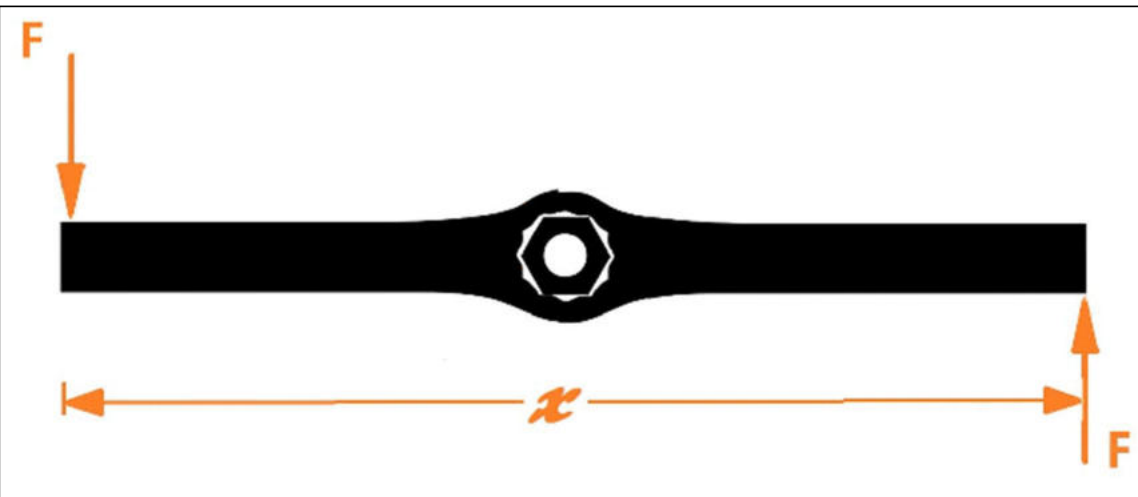


- Ve (Negative)



Definition : Couple

- ▶ Two force having the same magnitude and parallel line of action and of opposite sense are said to form a couple.
- ▶ Having same magnitude but opposite sense parallel to each other the sum of components in any direction is zero.
- ▶ Hence two force will not translate the body on which they act but tend to make it rotate.
- ▶ A moment is produce by two equal, opposite parallel & non-collinear forces is called Couple.



Magnitude of Couple

Moment (M) = Force (F) x arm (x)

$$M = F \times x$$

Characteristics of Moment & Couple

■ Moment

- It is produced by a single force not passing through C.G of the body.
- Moment = Force $\times$ $\perp$ distance
$$M = F \times l$$
- The force translates the body in the direction of force and rotates the body. It has resultant force.
- To balance the force causing moment, a collinear equal and opposite force is required.
- It is an example of fixed vector.

■ Couple

- It is produced by two equal and opposite parallel, non-collinear forces.
- Couple = Force $\times$ lever arm
$$M = F \times x$$
- Resultant force of couple is zero. Hence, body does not move. But rotate only.
- Couple can not be balanced by a single force. It can be balanced by a couple only.
- It is an example of free vector.

Varignon's Theorem

- “The algebraic sum of the moments of forces about any point in their plane is equal to the moment of their resultant about that point.”

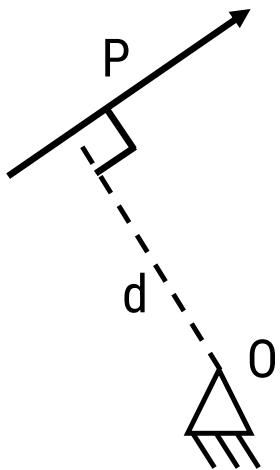


Figure (a)

Moment at “O” Point in figure (a)

$$M_O = P \times d$$

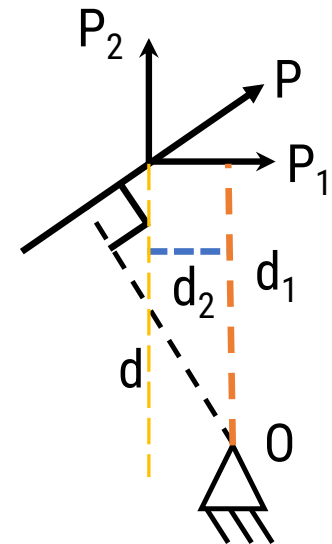


Figure (b)

Moment at “O” Point in figure (b)

$$M_O = P_1 \times d_1 + P_2 \times d_2$$

Characteristics of a Force

- ▶ It has a magnitude
- ▶ It has a direction
- ▶ It is a vector quantity
- ▶ It has a point of application
- ▶ It has a nature
 - ↳ Tensile Force
 - ↳ Compressive Force
 - ↳ Pull Force
 - ↳ Push Force

Resultant of Non-Concurrent Forces

► Summation of horizontal force

$$\sum F_x = F_1 \cos \theta_1 + F_2 \cos \theta_2 + \dots + F_n \cos \theta_n$$

► Summation of Vertical force

$$\sum F_y = F_1 \sin \theta_1 + F_2 \sin \theta_2 + \dots + F_n \sin \theta_n$$

► Magnitude of Resultant

$$R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$$

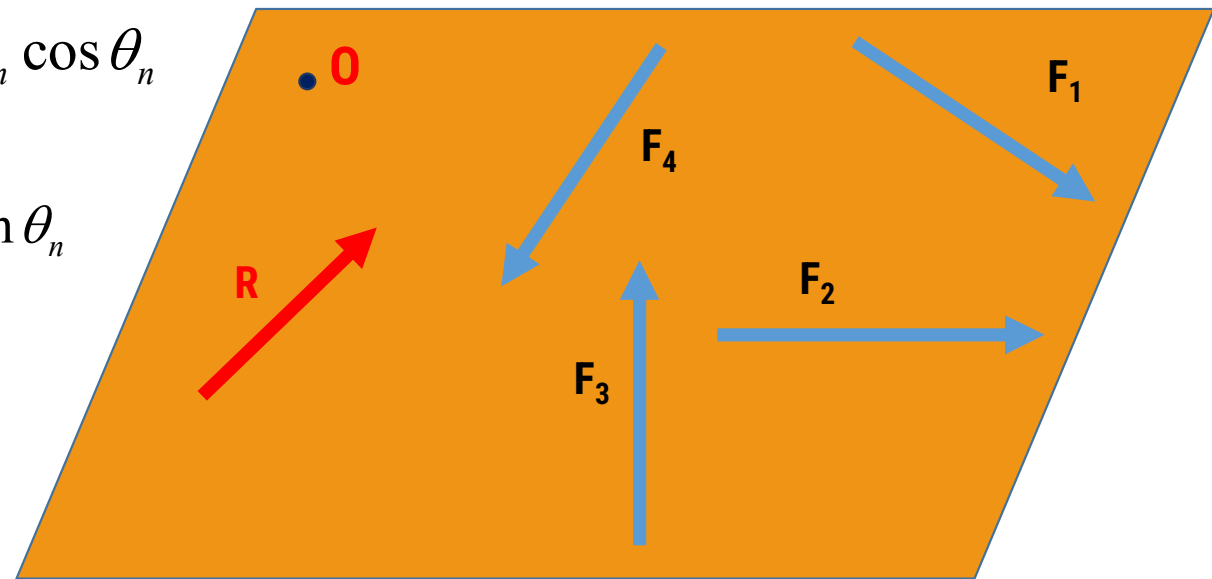
► Direction of Resultant

$$\tan \alpha = \frac{\sum F_y}{\sum F_x}$$

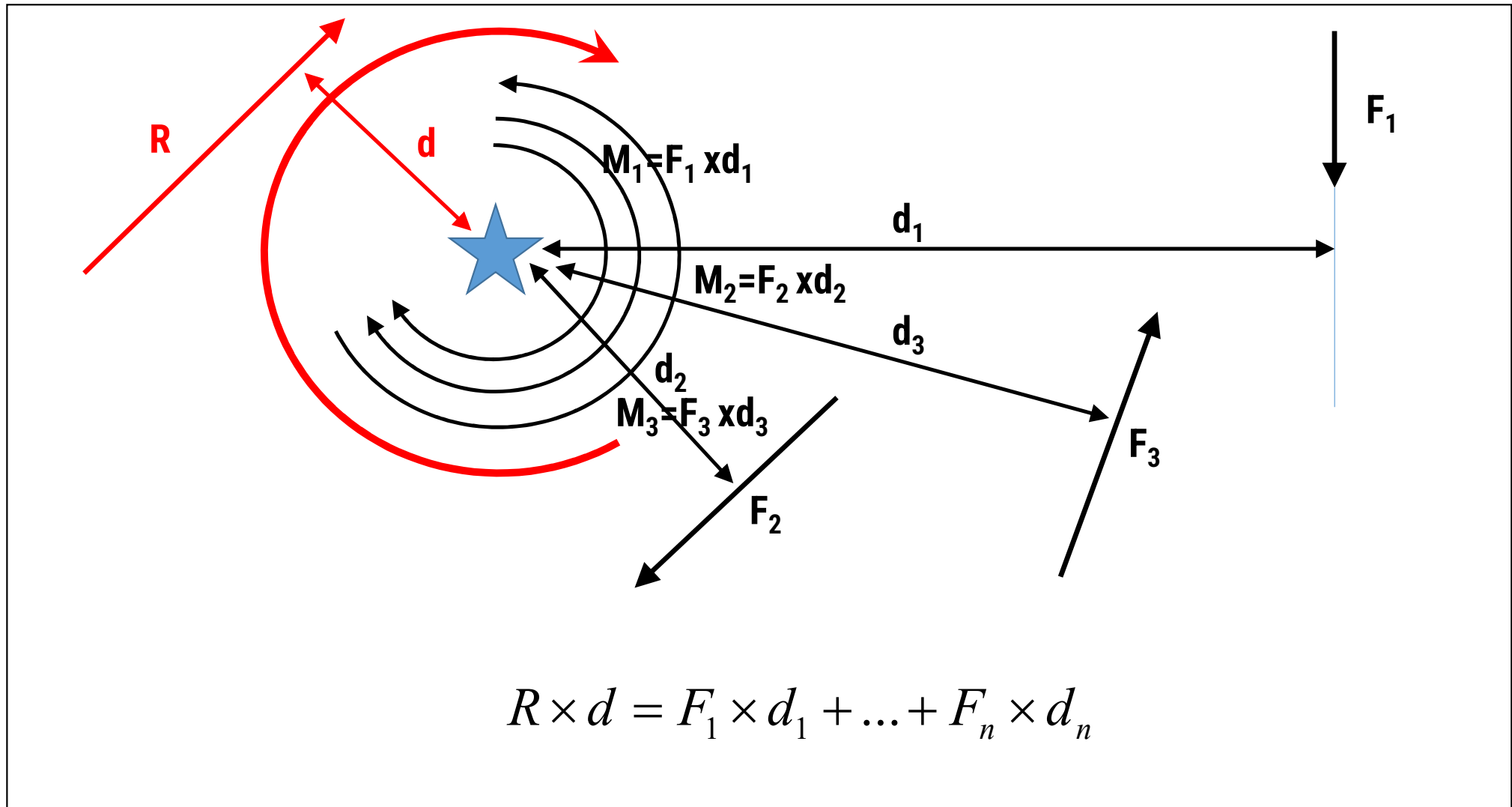
► Determine location of resultant

$$R \times d = F_1 \times d_1 + \dots + F_n \times d_n \Rightarrow$$

$$d = \frac{\sum (P \times d)}{R}$$

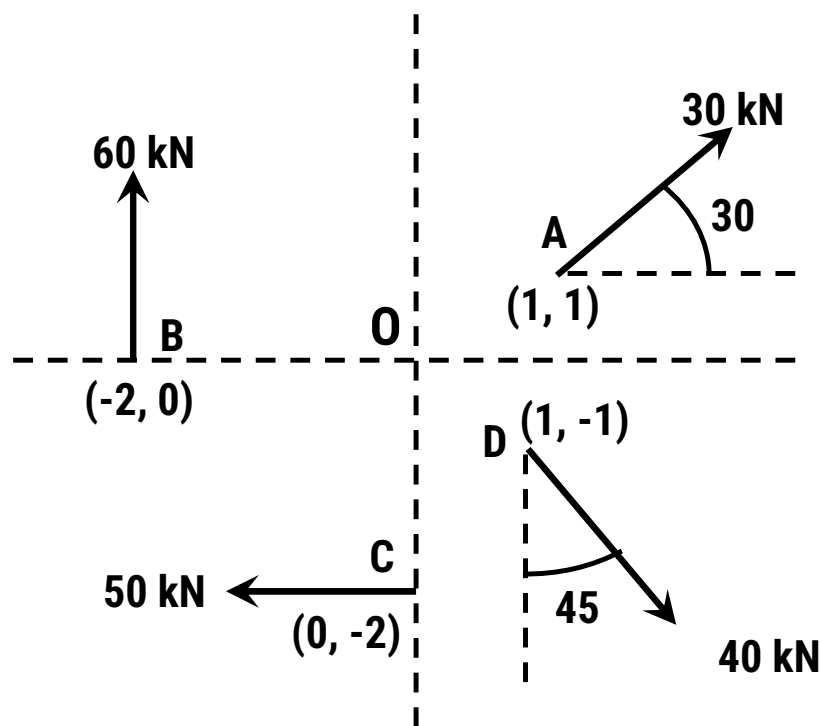


Non-Concurrent Coplanar Forces



Ex - 10

Find magnitude, direction and location of resultant of force system with respect to point 'O'



➤ **Answer**

From a given data

Force	Angle
$F_1 = 30 \text{ kN}$	$\theta_1 = 30$
$F_2 = 60 \text{ kN}$	$\theta_2 = 90$
$F_3 = 50 \text{ kN}$	$\theta_3 = 180$
$F_4 = 40 \text{ kN}$	$\theta_4 = 315$

➤ Summation of horizontal force

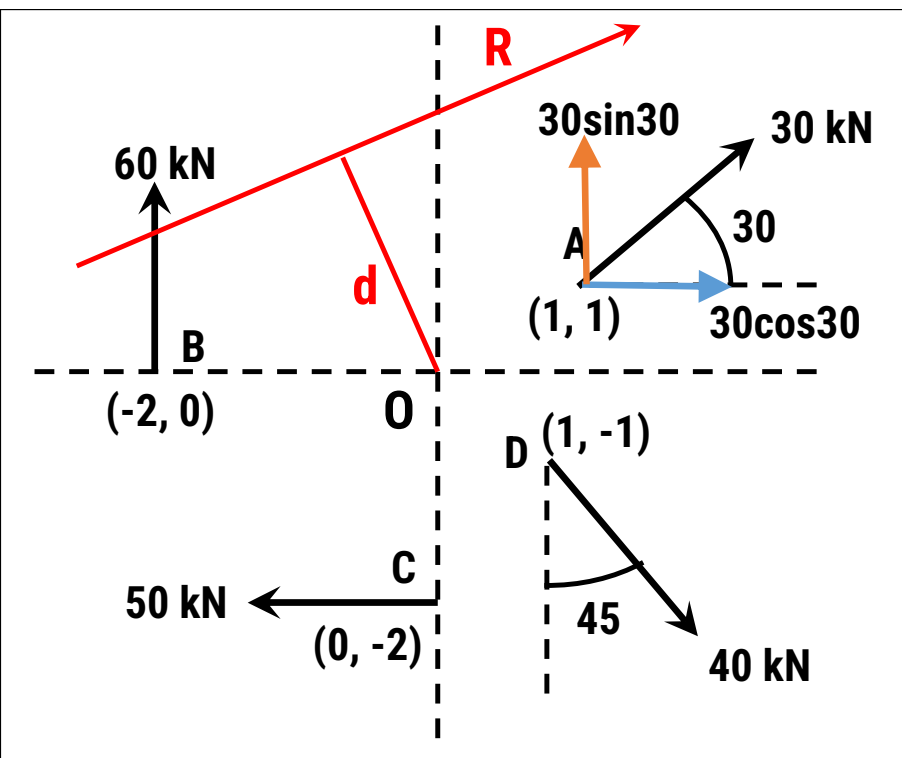
$$\sum F_x = 30 \cos 30 + 60 \cos 90 + 50 \cos 180 + 40 \cos 315$$

$$\sum F_x = 4.26 \text{ kN} \rightarrow$$

➤ Summation of Vertical force

$$\sum F_y = 30 \sin 30 + 60 \sin 90 + 50 \sin 180 + 40 \sin 315$$

$$\sum F_y = 46.72 \text{ kN} \uparrow$$



Force	Angle
$F_1 = 30 \text{ kN}$	$\theta_1 = 30$
$F_2 = 60 \text{ kN}$	$\theta_2 = 90$
$F_3 = 50 \text{ kN}$	$\theta_3 = 180$
$F_4 = 40 \text{ kN}$	$\theta_4 = 315$

► Magnitude of Resultant

$$R = \sqrt{(4.26)^2 + (46.72)^2}$$

$$R = 46.91 \text{ kN}$$

► Location of resultant

R will be in the first quadrant. Let, R be at perpendicular distance 'd' from 'O'

+Ve (Positive)

-Ve (Negative)

$$R \times d = (60 \times 2) + (30 \cos 30 \times 1) - (30 \sin 30 \times 1) + (50 \times 2)$$

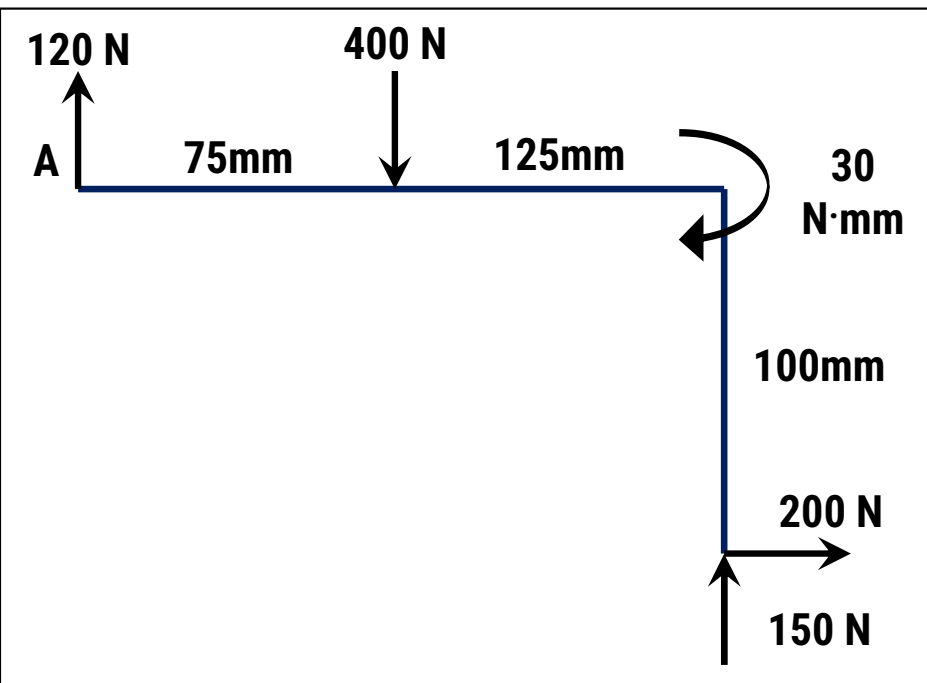
$$46.91 \times d = 120 + 25.98 - 15 + 100$$

$$d = 4.92 \text{ unit}$$

40 kN force is acting at 45° , hence its line of action passing through 'O'

Ex - 11

Find magnitude, direction and location of resultant of force system with respect to point 'A'



➤ **Answer**

From a given data

- ▶ Summation of Horizontal force

$$\sum F_x = 200\text{ N} \rightarrow$$

- ▶ Summation of Vertical force

$$\sum F_y = 120 - 400 + 150 = -130\text{ N} \downarrow$$

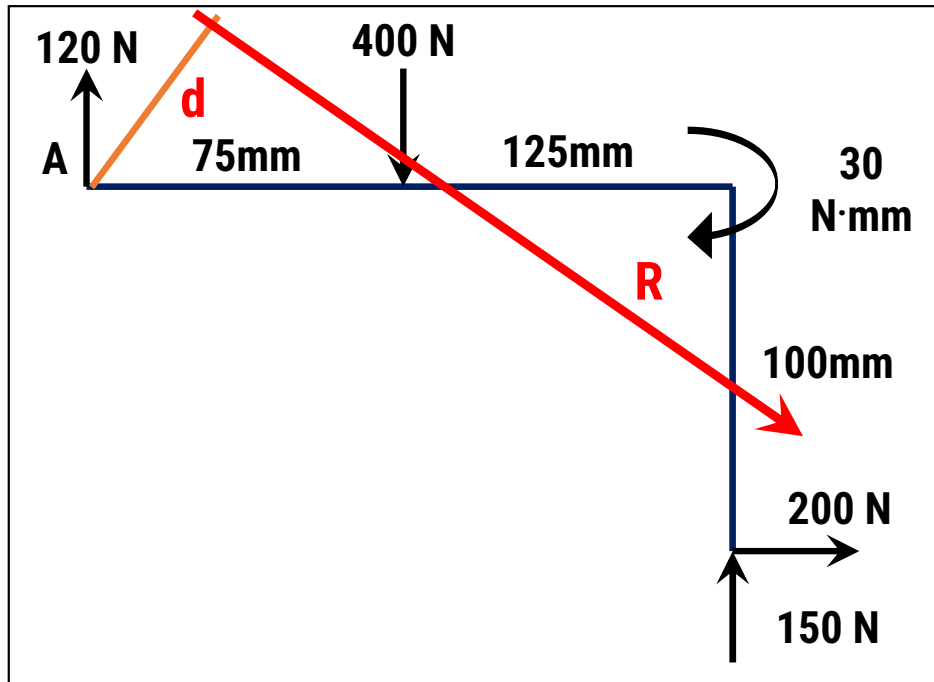
- ▶ Magnitude of Resultant

$$R = \sqrt{(200)^2 + (-130)^2} = 238.54\text{ N}$$

- ▶ Direction of Resultant

$$\tan \alpha = \left| \frac{-130}{200} \right| \Rightarrow \alpha = 33.02$$

- ▶ Here, $\sum F_x$ is positive and $\sum F_y$ is negative. Therefore resultant lies in fourth quadrant.



$$\Sigma F_x = 200\text{ N} \rightarrow \quad \Sigma F_y = -130\text{ N} \downarrow$$

$$R = 238.54\text{ N} \quad \alpha = 33.02^\circ$$

► Location of Resultant

Applying varignon's theorem

$$R \times d = \Sigma(F_x \times y) + \Sigma(F_y \times x)$$

$$+ R \times d \curvearrowright$$

$$= 120 \times 0 + 400 \times 75 \curvearrowright + 30 \curvearrowright$$

$$- 200 \times 100 \curvearrowleft - 150 \times 200 \curvearrowleft$$

$$238.54 \times d = 400 \times 75 + 30 - 200 \times 100 - 150 \times 200$$

$$238.54 \times d = -19970 \implies d = \frac{-19970}{238.54} = -83.72\text{ mm}$$

Here, negative value of 'd' indicates that resultant is on other side of origin 'A'